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ChargeMate Ops
System Design Document

Assignment 2 • 1 November 2026
Group 3 Team A

Yousef Musleh — 48603090
Harrison McLuckie — 49154788
Cooper Went — 49067281
Isaac Lynn — 49067265

Requirements supplied by Group 3 Team B

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1. Project scope and requirements baseline

1.1 Purpose and intended audience

This SDD translates Team B's ChargeMate Ops requirements into architecture, data rules and planned verification for the Harbour North Plaza network. It supports implementation planning by developers and project managers; it does not report an implemented or tested system.

For A2, Group 3 Team A designs Team B's operator-facing Ops project. The earlier Customer SRS supplies external interface context, not the Ops baseline. [SRC-01, Section 4; SRC-04]

1.2 System scope and boundary

Authorised staff inspect stations/devices, record and resolve faults, view issues and record confirmed redistribution. Unknown, unavailable and conflicting information remain explicit. [SRC-01, Sections 5–6; SRC-05]

Ops covers station/device views, fault recording/resolution, recorded attention indicators and suitable devices' confirmed redistribution. Physical inspection/repair and station hardware remain external; planning a move is not completing it. The design assumes neither GPS nor continuous tracking. Customer accounts, payments, prices, penalties and the kiosk journey stay outside Ops. Exact integration ownership remains in Section 4.4; fault resolution alone does not establish rental eligibility.

The **ChargeMate Operator** is the primary actor; the **Rental Station System** and **Customer-Facing ChargeMate System** are external collaborators. The SRS establishes no separate maintenance role, so access uses action permissions pending GAP-04. [SRC-01, pp.5–6,14–15]

Customer expects Ops-side release/return confirmations, but Ops' FRs do not fully allocate them. C-INT provides the proposed exchange boundary while hardware-command ownership remains GAP-02. [SRC-03, pp.7–12,15,23–24]

1.3 Received baseline and source documents

Source ID	Source and version	Precise locator	Use in this design
SRC-01	Team B, *ChargeMate Ops Software Requirements Specification*, supplied as `SRS (1).pdf`; cover v1.2, 12 September 2026	Requirements/fit criteria pp.7–9; use cases pp.12–19; recorded open questions pp.30–37	Primary behaviour and acceptance baseline
SRC-02	Team B handover, `Untitled design (3).mp4`, duration approximately 8:01; recording date not established	Approximately 4:23–6:07: information quality, physical confirmation and unanswered questions; 6:23–7:49: more specific workflow claims	Explanation and discrepancy identification
SRC-03	Group 3 Team A, *ChargeMate Customer SRS*, `Group3_TeamA_COMP2050.pdf`; change log through v1.5 dated 13 September 2026	Dependencies and requirements pp.7–12; Customer–Ops Q&A pp.23–24; return/problem flows pp.17–19	External expectations and allocation conflicts
SRC-04	Supplied COMP2050 Assignment 2 instructions	System Design, RTM, Test Specifications and Staff Report headings	Deliverable structure and project swap
SRC-05	Released Project B brief, repository copy	Project B — ChargeMate Ops, Sections 2–4	Plaza context and operator/customer boundary

The supplied ChargeMate Ops SRS is the requirements baseline for this design. Additional or conflicting details in the handover video are recorded as clarification questions or clearly labelled design assumptions.

1.4 Requirement reference list

The inherited baseline contains FR-01–FR-08 and NFR-01–NFR-03. Exact requirements and fit criteria are in SRC-01, pp.7–9; Section 11 maps every requirement to use cases, components and tests. `FR-xx` always means Ops; Customer requirements use the prefix `Customer.FR-xx`.

1.5 Use-case baseline

Source use case	Design interpretation	Source
UC-01 — View Station Status	Open monitoring, inspect available station information, select a station and recognise unavailable/conflicting/attention conditions	SRC-01 pp.12–13; Figure 3 p.16
UC-02 — Track a Power Bank	Search a unique ID and inspect available status/location; distinguish missing ID from unknown location and conflicting observations	SRC-01 pp.13–14; Figure 4 p.17
UC-03 — Report and Manage a Faulty Power Bank	Identify device, validate/save a fault, preserve exclusion and later record a permitted resolution; failed updates do not count as success	SRC-01 pp.14–19; Figure 5 p.18
Supporting redistribution scenario	Select candidate/source/destination, revalidate suitability, confirm physical movement and commit the new recorded location	FR-07; overall model p.11 and historical UC-04 discussion pp.36–38
Supporting issue-view scenario	Inspect the attention indicators generated by recorded faults, unavailable stations and inconsistent information	FR-08 and UC-01/02/03 failure/alternative branches

FR-07 requires redistribution despite the absence of a fourth complete detailed use case. The supporting scenarios are design elaborations; return, fault resolution and rental readiness remain separate facts.

2. Project vision

ChargeMate Ops gives Harbour North Plaza staff a shared view of rental stations, power banks and issues requiring attention. Operators can inspect reliable status information and follow a reported fault through investigation and resolution across shifts. The design makes unknown data and failed saves visible and excludes unresolved faulty devices from rental availability and redistribution. Its intended outcome is coordinated operational work with clear records that staff can trust and hand over between shifts.

3. Assumptions, clarification gaps and design decisions

A **gap** is an unanswered source question; an **assumption** is a provisional planning condition; a **decision** selects and justifies an implementation approach. Neither assumptions nor decisions imply stakeholder approval.

3.1 Open clarification register

The register records unresolved source questions, their impact and provisional handling. Source labels are defined in Section 1.3; no later stakeholder response is claimed.

Gap ID	Unanswered question and source	Affected requirements	Provisional handling
GAP-01	Exchange fields, ownership and update timing? [SRC-01, pp.5,30,33,37]	FR-01–FR-03, NFR-02	Agree the contract; retain source/time and expose unavailable/conflicting data. No numerical refresh promise.
GAP-02	Who authorises rentals, releases devices and confirms returns? [SRC-03, pp.7–12,15,23–24; SRC-01, pp.7–9]	Cross-system allocation; FR-02, FR-03	Keep C-INT as the exchange boundary; do not assume payment or hardware-control responsibilities.
GAP-03	Charge/availability criteria, exact counts and return capacity? [SRC-01, pp.7–8,30–31,37; SRC-03, pp.8,23]	FR-02, FR-05, FR-07	Separate charge, presence, occupancy and faults. Missing eligibility does not establish suitability or zero stock.
GAP-04	Required statuses/fault fields and resolver permissions? [SRC-01, pp.31,37]	FR-01, FR-03, FR-04, FR-06, NFR-03	Use ASSUMP-03 fields and per-action permission checks; no invented roles, categories or priorities.
GAP-05	Unavailable destination, competing changes or failed recording after physical movement? [SRC-01, pp.8,32]	FR-07, NFR-02	Recheck suitability/version and require physical confirmation; retain contradictory movement evidence for investigation.
GAP-06	Automatic alert triggers, priority and resolution rules? [SRC-01, p.32]	FR-08	Show recorded issues; add automatic thresholds/notifications only after agreement.

Gap ID	Unanswered question and source	Affected requirements	Provisional handling
GAP-07	Video claims of real-time telemetry, dock blocking and restored availability versus the SRS? [SRC-02, 6:23–7:49; SRC-01, pp.18–19,33]	FR-01, FR-03, FR-05, FR-06, NFR-02	Use latest-available facts. Keep local fault exclusion; agree remote blocking separately. Resolution alone proves neither charge, presence nor eligibility.
GAP-08	Performance, availability, backup, retention and recovery targets? [SRC-01, pp.7–9]	NFR-01–NFR-03; operations	Specify mechanisms and qualitative checks; seek agreement on numerical targets.

Open details do not weaken confirmed FRs: FR-07 still requires confirmed movement and excludes unsuitable candidates; FR-08 still requires recorded-issue visibility.

3.2 Design assumptions

These provisional assumptions support planning without adding requirements attributed to Team B.

Assumption ID	Provisional assumption	Reason for using it	Risk if incorrect / required change	Affected area and validation status
ASSUMP-01	Staff have a supported browser and network for normal tasks.	Common interface without separate installed apps.	Offline/restricted-device needs would change interface/deployment.	NFR-01/03; context unconfirmed
ASSUMP-02	An agreed setup process supplies registered station/device IDs and authorised staff identities.	SRS presupposes registration/access, not administration screens. [SRC-01, p.6]	Agree provisioning/import before implementation.	FR-01–FR-07, NFR-03; setup open
ASSUMP-03	Faults need a registered device ID and non-empty description; resolutions need a note and permitted operator.	Concrete provisional validation for valid device-linked records.	Revised fields require coordinated form, validation and test changes.	FR-04/06, NFR-03; fields/access open
ASSUMP-04	Multiple faults may coexist; any unresolved fault keeps the device excluded.	One resolution cannot hide another issue.	A one-active-fault policy would change schema/workflow, while retaining exclusion.	FR-04–FR-06; policy unconfirmed
ASSUMP-05	Sources supply stable event IDs and ordering information where automated deduplication/ordering is required.	Recognise retries and late data independently of arrival time.	Without comparable ordering, retain contradictions for investigation.	FR-01–FR-03, NFR-02; contract open

No charge threshold, refresh deadline, low-stock trigger or recovery target is invented. Unique IDs do not imply GPS: location may remain unknown or require investigation.

3.3 Proposed design decisions

Drafted 1 November 2026, these choices remain proposals where confirmation is open. Their implications appear in Sections 4 and 9–12; the date is not evidence of a meeting or approval.

- **DEC-01:** Web UI, modular backend and PostgreSQL, rather than independent business services, keep related updates atomic with clear responsibilities. [FR-04–FR-07; NFR-01; ASSUMP-01]
- **DEC-02:** Separate fault, location, charge/occupancy and quality instead of one status flag: location or resolution cannot establish full eligibility. Dictionary remains GAP-03/04. [FR-02/03/05/06; NFR-02]
- **DEC-03:** Check backend action permissions and versions at save, rather than trust UI controls/stale reads. Access assignments remain GAP-04; competing changes need GAP-05 handling. [NFR-03; FR-05/07]
- **DEC-04:** Commit an outgoing record with the business change, then deliver/retry separately rather than wait for a remote response. This preserves local durability and exposes pending synchronisation. Contract remains GAP-01/07. [FR-04–FR-06; NFR-02]
- **DEC-05:** Use database-backed reads without a separate business cache/offline write queue to limit stale copies/commands. Revisit with measured workload or offline needs. [NFR-02; FR-05/07; ASSUMP-01]
- **DEC-06:** Display recorded issues before adding automatic alerts/recommendations; defined rules must precede automation. [FR-08; GAP-06; SRC-05 Section 3]

Decision changes require coordinated diagrams, data rules, RTM and tests. Record actual responses/reviews with evidence and date before marking an open item agreed.

4. System architecture and rationale

4.1 Architecture overview

The three layers are C-UI, a modular backend and C-DB (PostgreSQL). Staff monitor, search, manage faults and confirm redistribution through C-UI; the backend checks permission and business rules before accessing persistent records. [DEC-01; ASSUMP-01]

Backend modules form one application, allowing related fault/device/movement changes to commit atomically. C-INT routes external input through the same domain rules and delivers outgoing updates separately.

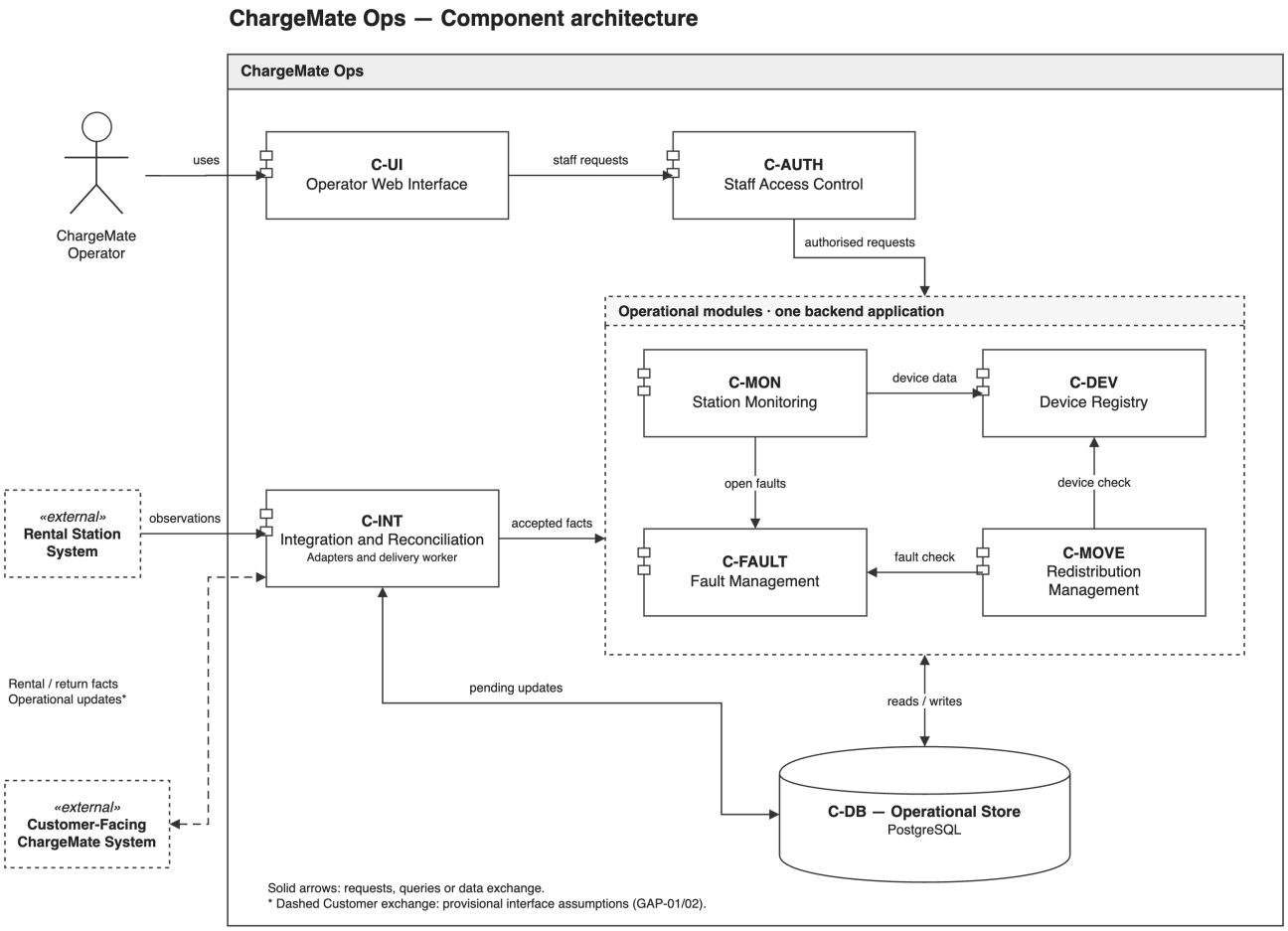


Figure ARCH-01. *Ops contains C-UI/C-AUTH, four domain modules, C-INT and C-DB. Operator, station and Customer systems are external. Solid arrows show logical exchanges; the dashed Customer exchange has a provisional contract. The figure assigns neither payments nor automatic hardware control. Editable source: ARCH-01.drawio, using draw.io.*

A modular backend fits the pilot's shared transaction needs. It trades independent business-service scaling/releases for simpler consistency and a shared application/database dependency.

4.2 Component catalogue

C- IDs are proposed component labels derived from the SRS functions. Use them consistently in architecture, RTM and tests; state diagrams may name the objects those components manage.

Component	Responsibility	Main exchanges	Requirements supported
C-UI — Operator Web Interface	Station/device views, fault/resolution forms, movement confirmation and qualified recovery messages.	Actions → backend; results/quality → screens	FR-01–FR-08, NFR-01, NFR-02

Component	Responsibility	Main exchanges	Requirements supported
C-AUTH — Staff Access Control	Identify staff and check each action/resource; deny when permission is unestablished.	Identity/action/target → allow/deny	NFR-03; all protected operational functions
C-MON — Station Monitoring	Combine station observations, device/fault facts and attention indicators.	Station query → summary/detail via C-DEV/C-FAULT	FR-01, FR-02, FR-08, NFR-02
C-DEV — Device Registry	Registered-ID lookup, location/facts and record versions.	Device ID → details, not-found or investigation	FR-03; identification/location support for FR-02, FR-04–FR-08
C-FAULT — Fault Management	Validate/save faults and resolutions; retain exclusion while any fault is unresolved.	Device-linked input → saved result or rejection/failure; exclusion → domain modules	FR-04, FR-05, FR-06, FR-08
C-MOVE — Redistribution Management	Check registered stations/current suitability and record confirmed movement.	Device/stations/physical confirmation → destination or conflict/failure	FR-07; affects FR-02, FR-03, FR-05
C-INT — Integration and Reconciliation	Validate source/message, deduplicate/order where supported, preserve conflicts and deliver updates.	External records → domain rules; outgoing updates → agreed recipient	FR-01–FR-03, FR-08, NFR-02; agreed fault/status exchange
C-DB — Operational Store	Persist stations/devices, faults, movements, observations, versions and processing records with constraints.	Protected transactions → commit/rollback	Persistent support for FR-01–FR-08 and NFR-02

C-MON queries device/fault data; C-MOVE checks C-DEV/C-FAULT before completion. C-FAULT/C-MOVE coordinate through the same device/version. Shared storage never bypasses domain rules.

External systems cannot access C-DB directly. C-INT validates service access and routes domain changes; its worker records delivery status without changing eligibility independently. Storage and concurrency rules follow in Sections 9–10.

4.3 Requirement-driven architecture rationale

NFR-01 drives visible C-UI task controls without database/CLI access, verified with representative users. NFR-02 drives C-INT source/time evidence and explicit quality in C-MON/C-DEV; missing values never become confirmed zero and timestamps promise no refresh deadline. NFR-03 drives backend per-action checks, default denial and restricted service access; hiding a button cannot protect records. Permission assignments remain GAP-04.

For FR-04–FR-06, shared C-FAULT/C-DEV transactions prevent partial saves and preserve exclusion while any fault remains unresolved. FR-07 drives save-time suitability, version and physical-confirmation checks: selecting a destination does not establish physical location. C-INT isolates provisional external contracts and conflicts under GAP-01/02/07.

These technical choices follow OWASP authorisation and PostgreSQL locking; those sources do not supply ChargeMate business rules.

4.4 External interface responsibilities

The proposed exchanges below support planning, not an existing API. Exact formats, ownership and timing remain open under Section 3.

Exchange	Direction and proposed information	Processing / failure behaviour	Responsibility and assumption
Station observations	Rental Station System → C-INT: station/device identifiers, observation time, available status/location/availability facts, event ID and source revision where supplied	Validate source and registration; preserve missing fields; record conflicts rather than overwrite contradictory facts blindly	Stations supply physical observations. Charge/capacity semantics and ordering support remain GAP-01/03 and ASSUMP-05.
Rental/return facts	Customer → C-INT: event ID, device ID, station ID where applicable, event type/time and source revision where available	Route accepted facts through C-DEV/domain rules; repeated or late input cannot silently erase newer or conflicting evidence	Customer supplies its recorded rental/return information. This exchange does not itself define who authorises release or confirms physical docking.

Exchange	Direction and proposed information	Processing / failure behaviour	Responsibility and assumption
Operational changes	C-INT → agreed Customer/station recipient: entity ID, changed operational facts, local revision and event ID	Save the pending outgoing update with the business change; retry delivery separately and distinguish pending from acknowledged delivery	Ops owns its recorded fault/movement changes. Remote exclusion/blocking requires the recipient's agreed handling; local success alone does not prove it happened.
Operational information requests	Agreed external caller → C-INT: permitted station/device query; response includes accepted values and quality/source/time information	Apply service access checks; return unknown/unavailable/conflicting information explicitly	Share only the fields needed by the agreed interface. Exact-count/return-capacity ownership remains GAP-01/03.

Docking/release hardware stays with the external station system; accounts, pricing and payments stay with Customer. Ops owns operational records/views. Command/acknowledgement ownership remains GAP-02; exchanges need no payment credentials.

Consistent command results

Protected writes carry a target ID, expected version and operation ID. Outcomes are **committed**, **validation rejected**, **not found**, **access denied**, **version conflict** or **save failed**. Commit returns the saved ID/version. An authorised retry with the same identity, operation ID and input returns that result without repeating the change; altered reuse is rejected. Rejected/rolled-back requests never leave success receipts.

5. User experience: two wireframe/storyboard sets

5.1 Storyboard SB-01 — UC-01 View Station Status

Actor: An authorised ChargeMate Operator checks the plaza network at shift start or busy periods, using a desktop or tablet (ASSUMP-01). Visible controls must support station checks without database or command-line access (NFR-01).

Preconditions: The operator is signed in and authorised to view station information (NFR-03). The stations are registered in ChargeMate Ops (SRC-01 A01).

Requirements: FR-01, FR-02, FR-05, FR-08, NFR-01, NFR-02, NFR-03

SB-01 Storyboard sequence

Step	Operator action / screen	System response and next step
1	Open station monitoring (WF-01).	Show station identities and available status/availability; qualify missing information.
2	Select a station (WF-02).	Show that station's recorded details and last-update information.
3	Inspect an attention or data-quality condition (WF-03).	Distinguish recorded faults from unavailable/conflicting data; retain source evidence for investigation.
4	Retry after a failed retrieval.	Replace qualified values only after reliable information arrives; a failure is never displayed as a confirmed zero.

SB-01 Wireframes

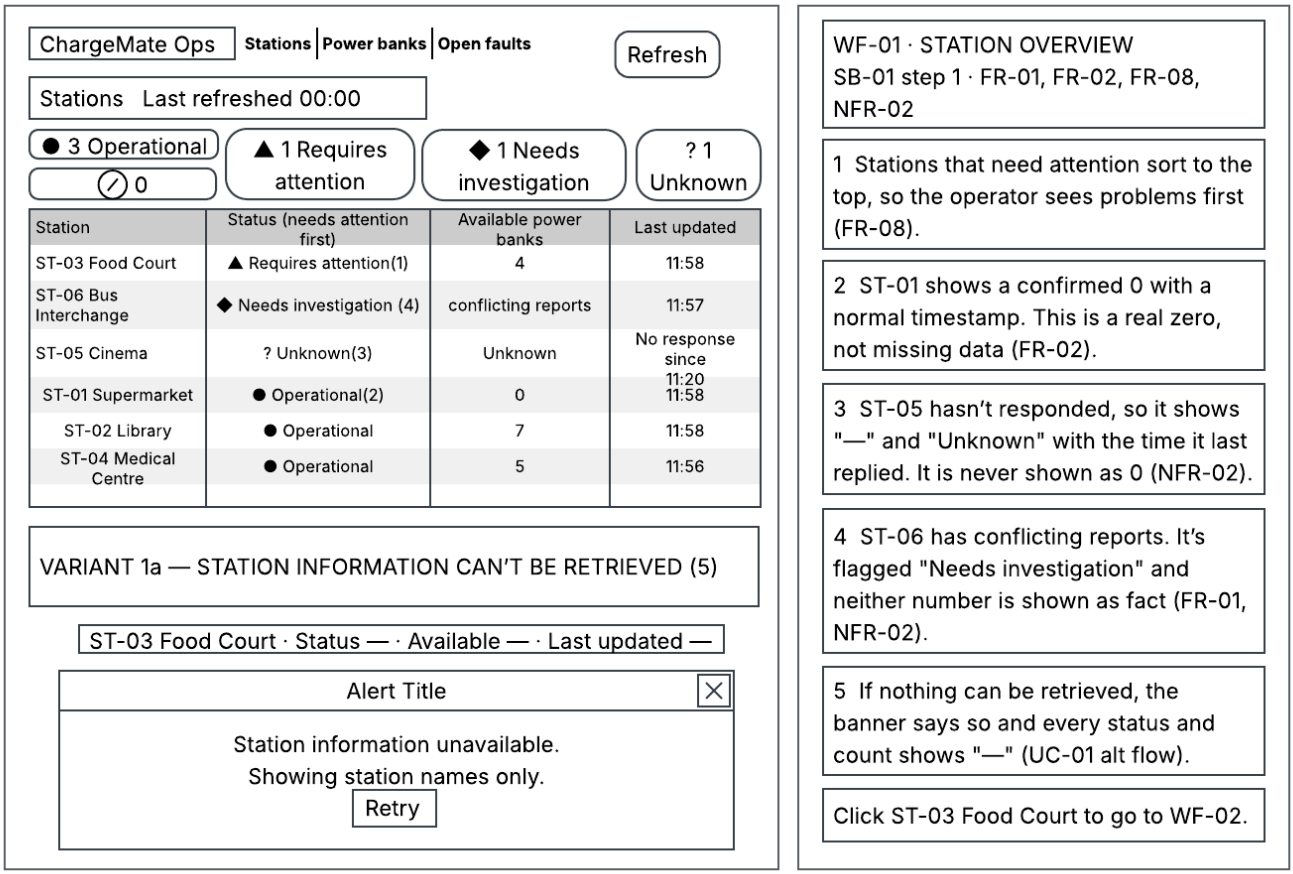


Figure 5.1.1 — WF-01 Station overview.

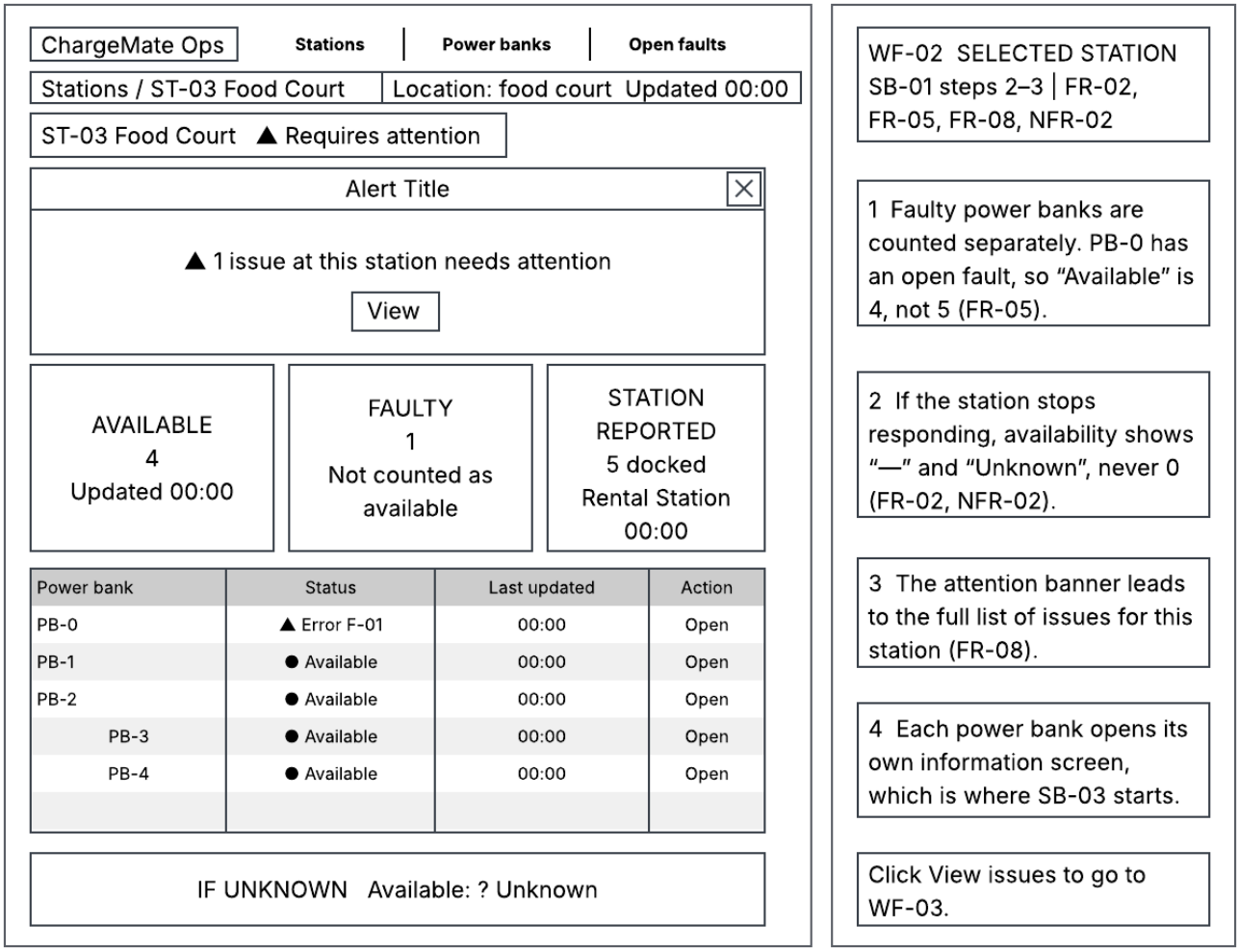


Figure 5.1.2 — WF-02 Selected station.

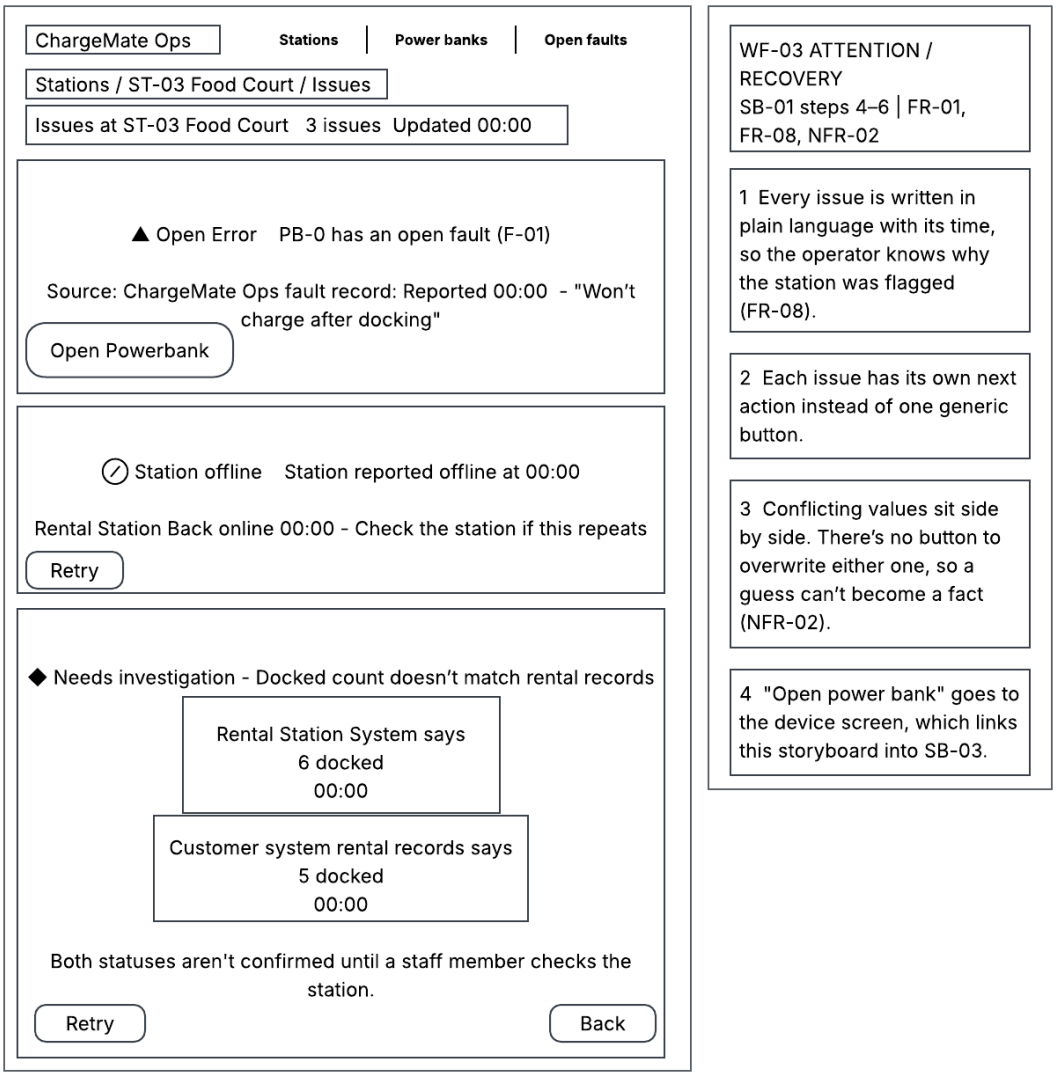


Figure 5.1.3 — WF-03 Attention / recovery view.

5.2 Storyboard SB-03 — UC-03 Report and Manage a Faulty Power Bank

Actor: The same operator reports a damaged/non-charging device, often at its station. Another shift may later resolve it. The short form supports device-side reporting; resolution is an action permission, not an invented separate role. [SRC-01 Section 7.3.3; GAP-04; ASSUMP-03]

Preconditions: The operator is signed in and has permission to record faults (NFR-03). The power bank is registered with a unique ID (SRC-01 A02).

Requirements: FR-03, FR-04, FR-05, FR-06, FR-08, NFR-01, NFR-02, NFR-03

SB-03 Storyboard sequence

Step	Operator action / screen	System response and next step
1	Search the power-bank ID (WF-04).	Match a registered device; distinguish not-found from unknown location.
2	Enter a fault description (WF-05).	Validate required details and action permission. Missing details keep the operator on the form.
3	Submit and inspect the result (WF-06).	Confirm only a committed save; retain input on rejection. A lost response is unconfirmed and is checked using the same operation ID.
4	Later inspect the unresolved fault (WF-07).	Preserve its device association, attention indication and exclusion while unresolved.
5	An authorised operator records a resolution (WF-08).	Keep the fault active after a failed save; after success retain resolution history. Other unresolved faults still exclude the device, and charge/location eligibility is checked separately.

SB-03 Wireframes

Supporting screens: WF-04 Device search/detail and WF-07 Later fault review. The complete sequence is specified above; the embedded samples below show fault entry, save outcomes and resolution.

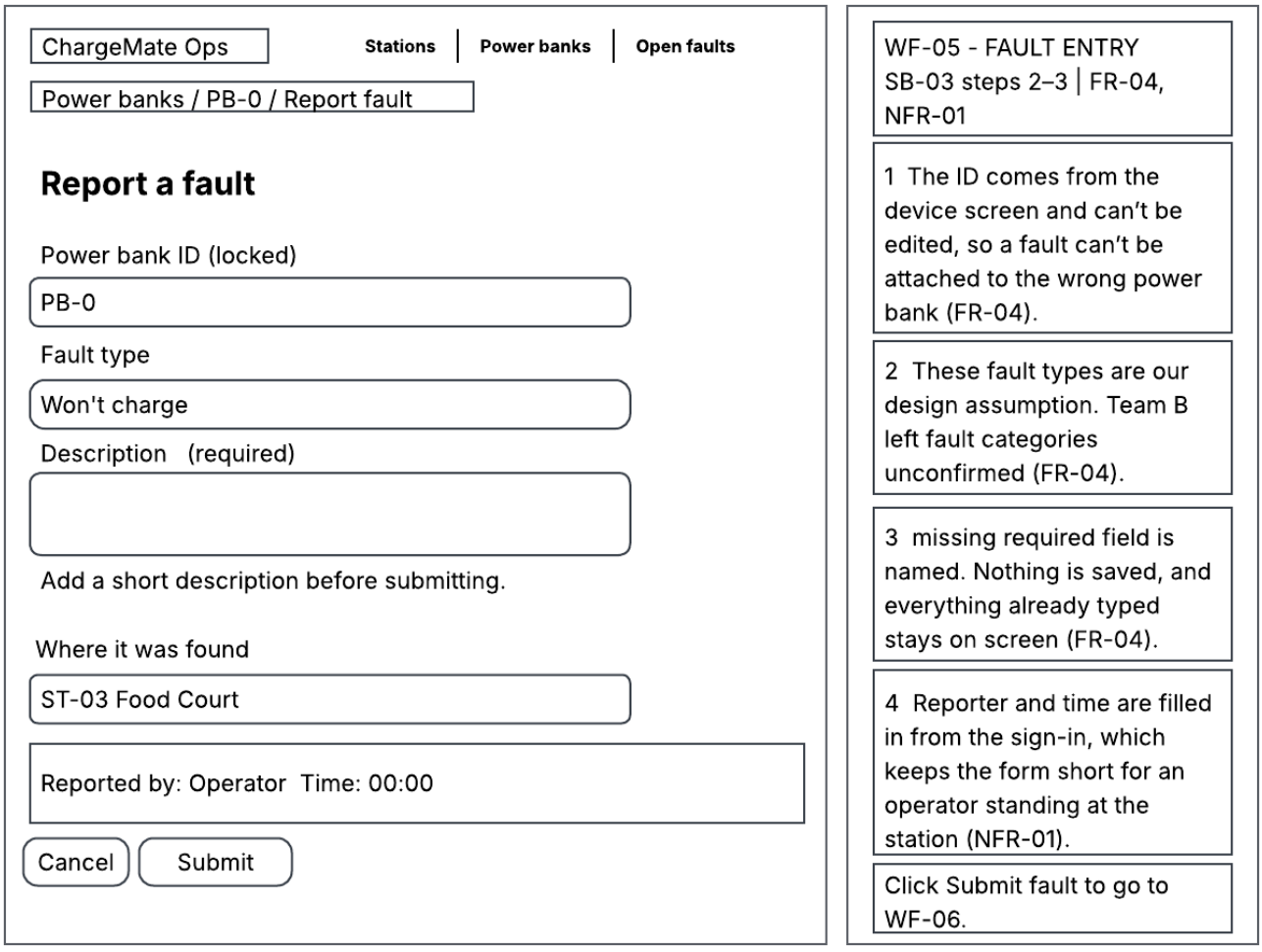


Figure 5.2.1 — WF-05 Fault entry.

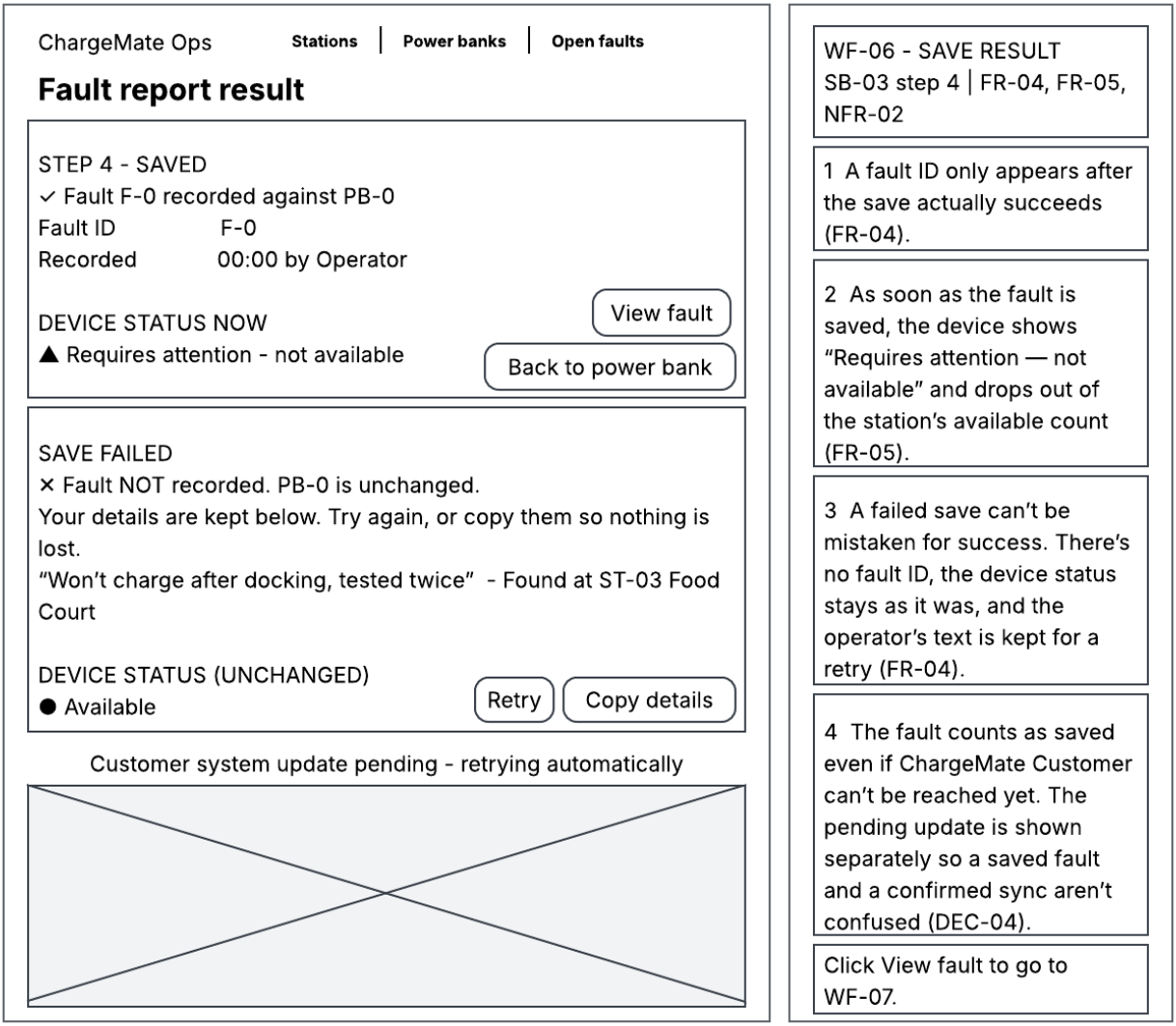


Figure 5.2.2 — WF-06 Save result.

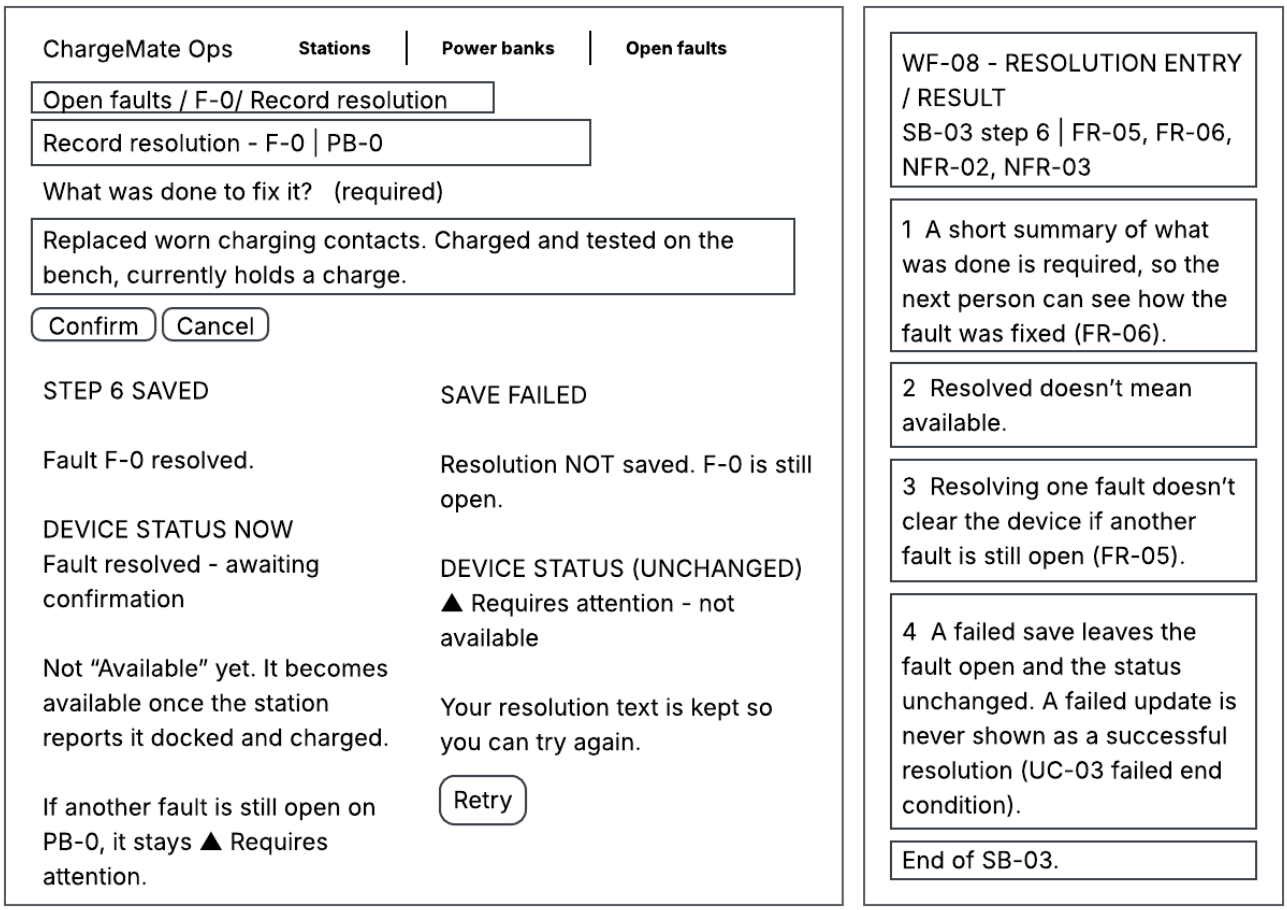
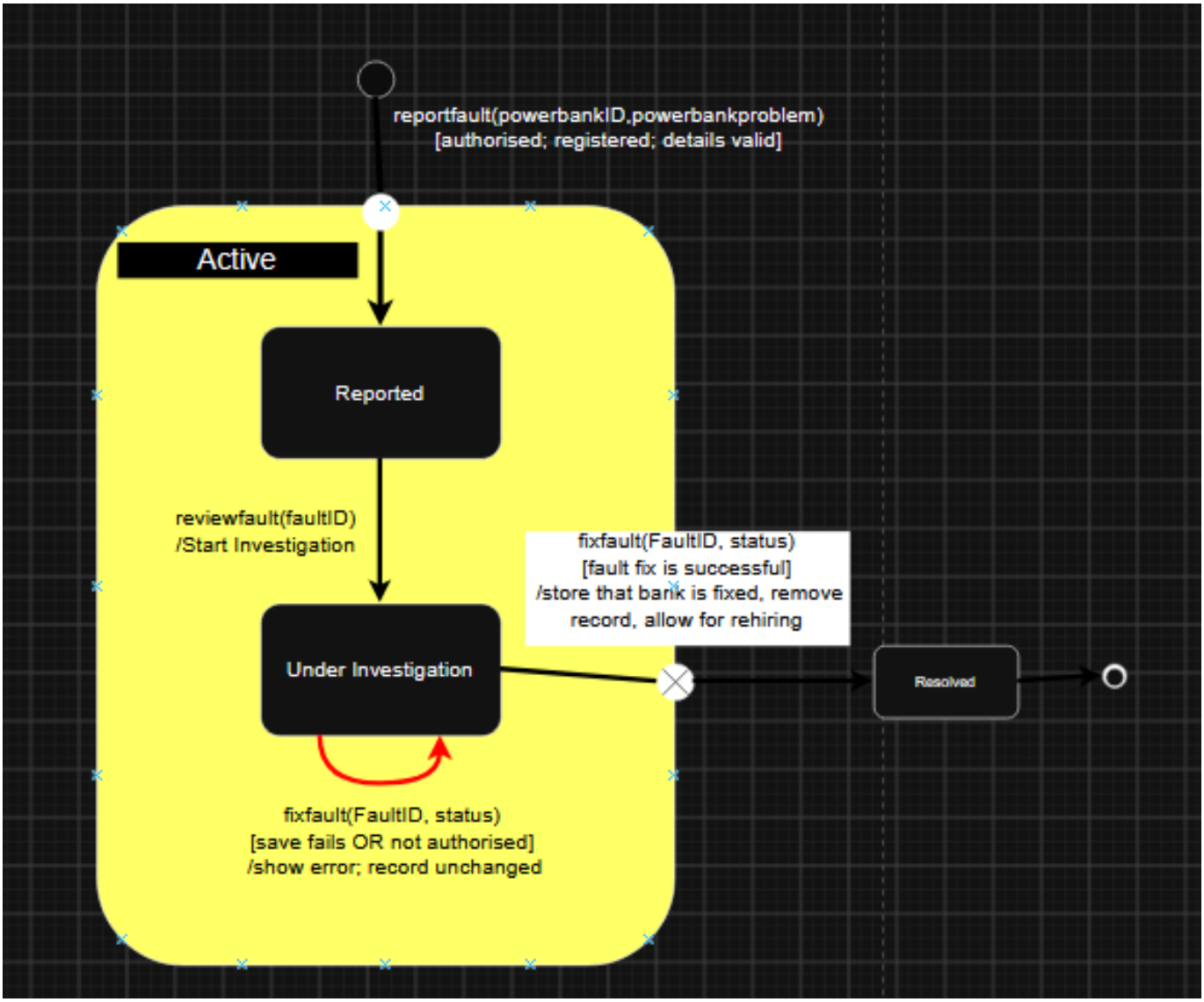


Figure 5.2.3 — WF-08 Resolution entry / result.

6. State diagram 1 — FaultRecord

Object and purpose: FaultRecord Represents a reported fault against a submitted power bank. It tracks when the faulty record was submitted, and what is wrong with it. **Requirements/use cases:** UC-03 – Report and Manage a Faulty Power Bank, FR-04, FR-05, FR-06, FR-07, FR-08 **Figure SD-01:**



Resolution closes the active fault workflow while retaining the stored record/history. The figure's rental-availability wording is subject to the independent eligibility checks in Sections 9 and 12; resolution alone does not certify rental readiness.

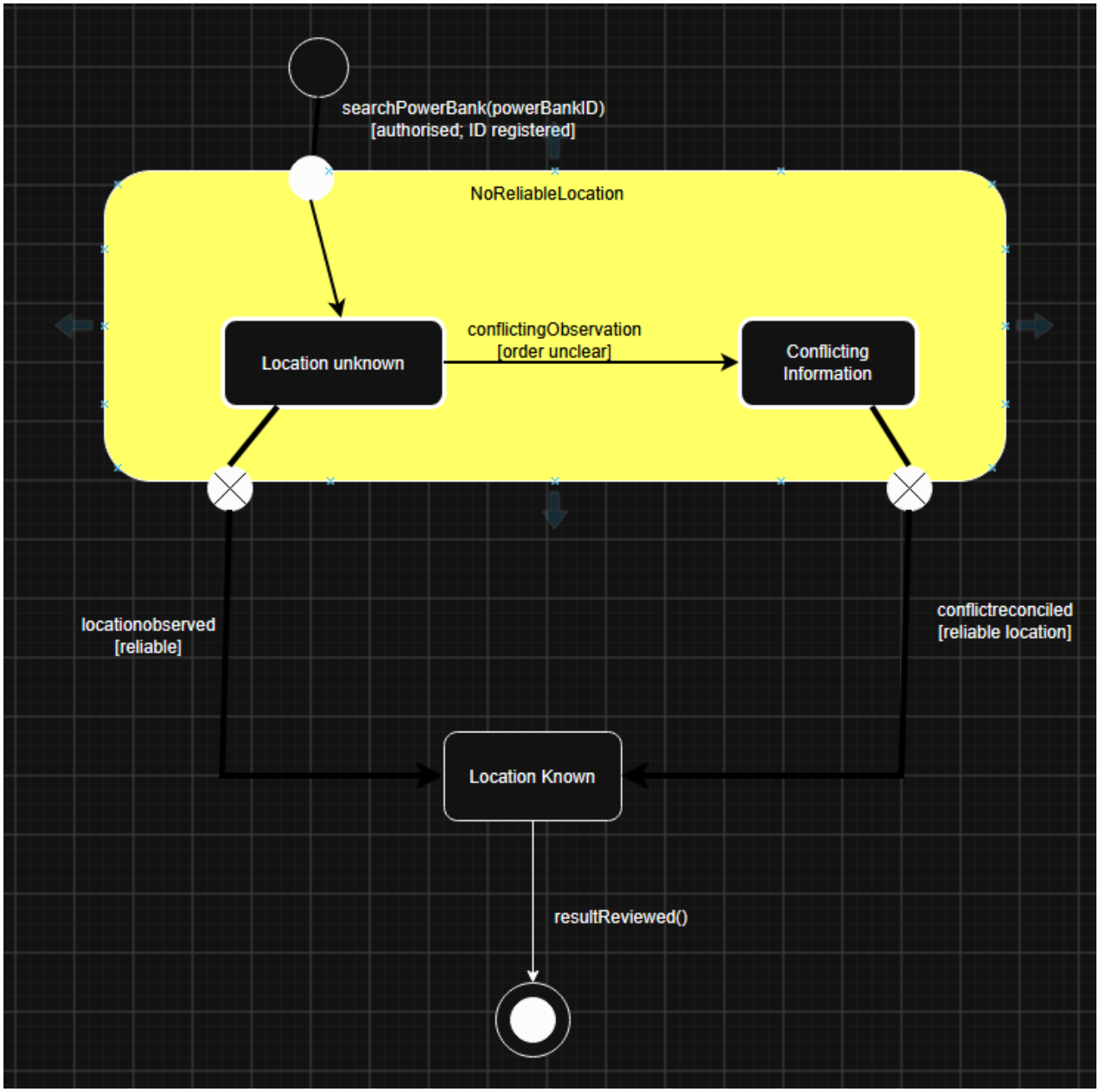
Current state	Event	Guard / condition	Action	Next state	Related test
Initial	reportfault(powerbankID, powerbankproblem)	Operator authorised; device registered; details valid	Create record for the power bank, mark it as faulty	Reported (inside Active)	T-09, T-10, T-11, T-18, T-23
Reported	reviewfault(FaultID)	fault exists	record that the fault is under investigation	Under investigation	T-12
Under investigation	fixfault(FaultID, status)	Authorised resolution; valid note; save commits	retain resolution note/actor/time and device history; recompute unresolved-fault exclusion without certifying rental eligibility	Resolved	T-12, T-13
Under investigation	fixfault()	save fails	show error, leave record unchanged	Under investigation	T-14
Under investigation	fixfault()	reviewer has improper access	show error, leave record unchanged	Under investigation	T-14, T-18

Current state	Event	Guard / condition	Action	Next state	Related test
Resolved	workflowClosed()	Resolution already committed	Close workflow; retain record/history	Final	T-12 (retention); closure is a model review check

Valid Creation: Only an authorised person can report that a fault has been created/fixed. it is then flagged as needing repairs, and that the repair is active. **Unresolved Investigation**: The record stays active when an error occurs, leaving the data untouched so it can be successfully resolved **Successful Resolution**: The authorised user submits the fixed fault, and the save succeeds. the records are resolved and the device status is updated to reflect this. **Failed Save**: If the save fails, the record of the fault remains active, and no further data is saved. The failure isnt treated as a resolution to the fault. **Unauthorised Resolution**: the resolution is denied, and the record remains active.

7. State diagram 2 — PowerBank Tracking Status

Object and purpose: Powerbank tracking status. represents the data that ChargeMate Ops can see about the location of a power bank, separating known location, and information that needs to be investigated.
Requirements/use cases: UC-02 - Track a Power Bank, FR-03 **Figure SD-02:**



Current state	Event	Guard / condition	Action	Next state	Related test
Initial	searchPowerBank(powerBankID)	Operator authorised; ID registered; information retrievable	Open tracking lookup; display "unknown" until reliable information is available	No reliable location → Location unknown (via entry point)	T-06, T-07
Location unknown	locationObserved(id, location, source, time)	Observation is reliable and consistent with recorded data	Display recorded status and location	Location known (via exit point)	T-06
Location unknown	conflictingObservation(id)	Sources disagree and order cannot be established (ASSUMP-05)	Keep both observations; display "requires investigation"	Information conflicting	T-08
Information conflicting	conflictReconciled(id)	A reliable location is established	Record outcome; clear the investigation flag; display location	Location known (via exit point)	T-08
Location known	resultReviewed()	Operator has viewed the information	Close the lookup	Final	T-06

What the Object is: Diagram shows 1 pass of the location tracking on the power bank **How the Lookup Starts:** an authorised operator starts the search with an id for a powerbank. Location is put into location unknown state until a valid location is found. **Failed Search:** If the id is not registered or cannot be retrieved, an error is shown and nothing is stored. **Unknown State:** If the operator cannot see the current state of the power bank, a specific unknown value is returned, instead of a guessed value. **Conflicting Information:** if multiple sources of data come in and the order is not proper, the operator will see a requires investigation flag to try and fix it. **Successful Outcome:** in order for the outcome to be successful, a meaningfully ordered location data must be transmitted and saved to the main operator **Unchanged on failure:** if something fails, the system saves no data and nothing is altered.

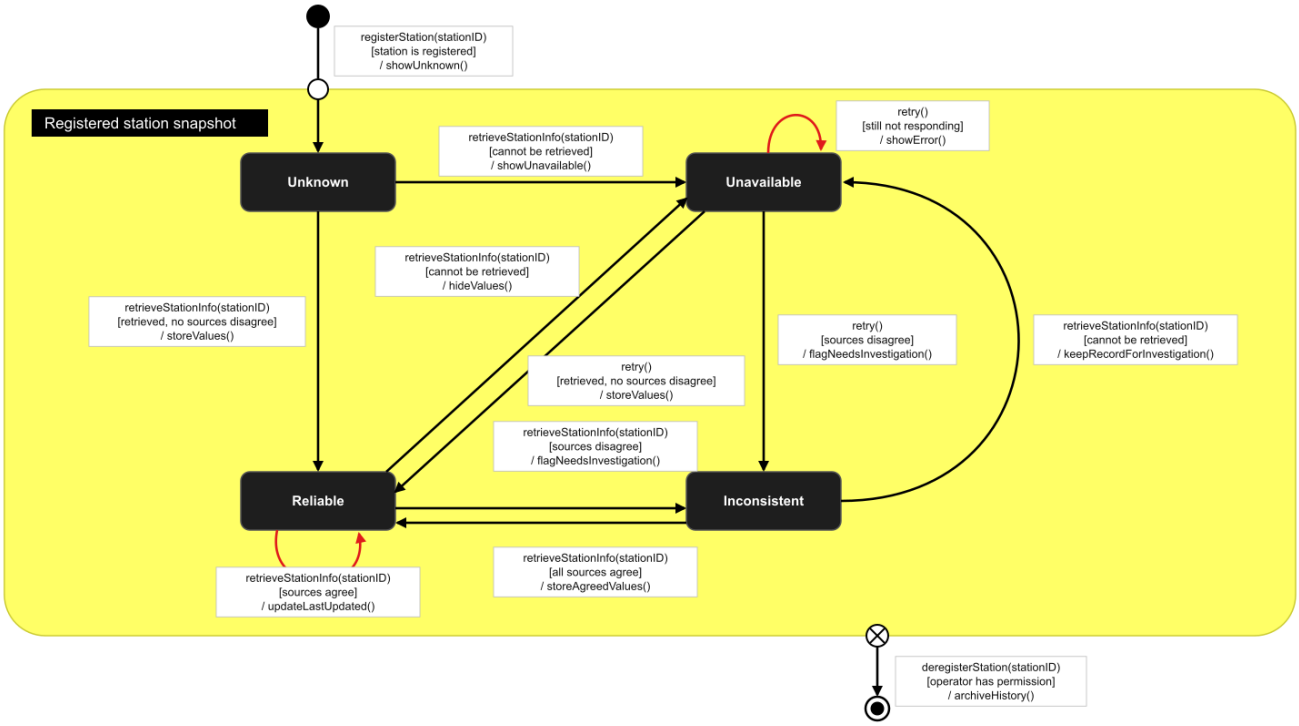
8. State diagram 3 — OperationalSnapshotQuality

Object and purpose: OperationalSnapshotQuality represents how trustworthy the information ChargeMate Ops currently holds about a rental station is. It does not track whether the station itself is working. A station can be reported as offline and that report can still be reliable. It only tracks whether the system can trust the status and availability it is showing the operator in UC-01.

Requirements/use cases: UC-01 – View Station Status, FR-01, FR-02, FR-08, NFR-02

Related screens: WF-01, WF-02, WF-03

Figure SD-03:



Current state	Event	Guard / condition	Action	Next state	Related test
Initial	registerStation(stationID)	station is registered	create a snapshot with no values, show "Unknown"	Unknown	T-01
Unknown	retrieveStationInfo(stationID)	information retrieved, no sources disagree	store status and availability with source and time, show "Last updated"	Reliable	T-01
Unknown	retrieveStationInfo(stationID)	information retrieved but sources disagree	preserve both observations and their source/time; flag “Needs investigation”	Inconsistent	T-03
Unknown	retrieveStationInfo(stationID)	information cannot be retrieved	show "Unavailable" and "—" instead of a number	Unavailable	T-02
Reliable	retrieveStationInfo(stationID)	information retrieved, sources agree	update values and "Last updated" time	Reliable	T-01
Reliable	retrieveStationInfo(stationID)	Rental Station System and Customer system disagree	keep both values with their source, flag "Needs investigation"	Inconsistent	T-03
Reliable	retrieveStationInfo(stationID)	information cannot be retrieved	hide values, show "Unavailable" with the time of the last response	Unavailable	T-02
Unavailable	retry()	information retrieved, no sources disagree	store values, remove "Unavailable" label	Reliable	T-02
Unavailable	retry()	information retrieved but sources disagree	keep both values, flag "Needs investigation"	Inconsistent	T-03
Unavailable	retry()	station still not responding	show error, leave values hidden	Unavailable	T-02
Inconsistent	retrieveStationInfo(stationID)	all sources now agree	store agreed values, remove "Needs investigation"	Reliable	T-03
Inconsistent	retrieveStationInfo(stationID)	information cannot be retrieved	show "Unavailable", keep record for investigation	Unavailable	T-03
Any state	deregisterStation(stationID)	operator has permission	archive the snapshot history	Final state	T-26

Initial conflicting retrieval: The Unknown → Inconsistent transition in the table also applies when the first successful retrieval already contains conflicting observations. The table specifies this path; it is not explicit in the supplied figure. No conflicting count is shown as confirmed.

- **Unknown information**: The snapshot starts as Unknown because nothing has been retrieved for the station yet. The system shows "Unknown" and "—", never 0, so a station with no data can't be mistaken for an empty station.
- **Reliable information**: The latest station information was retrieved and no source disagrees with it (UC-01 main success scenario). Status and availability are shown with the time they were last updated.
- **Unavailable information**: The station information couldn't be retrieved (UC-01 alternative flow "Station Information Unavailable"). The system hides the old values instead of showing them as if they were current, and shows when the station last responded (NFR-02).
- **Inconsistent information**: The sources disagree, for example the Rental Station System says 6 docked but the rental records imply 5 (UC-01 alternative flow "Inconsistent Station Information"). Both values are kept side by side and the station is flagged "Needs investigation". Neither value is treated as confirmed and the operator can't overwrite either one (WF-03).
- **Quality vs health**: This only describes how much the station information can be trusted. Whether the station is offline or has a faulty power bank is stored separately (DEC-02).
- **No deadline**: The SRS doesn't give a time limit for how old station information can be (GAP-01), so it only moves to unavailable when it actually fails, not after a set amount of time.

9. Persistent data, caching and maintenance

9.1 Persistent data strategy

C-DB is server-side PostgreSQL storage for durable, linked station/device/fault/movement records. It supports shared transactions under DEC-01–DEC-04; PostgreSQL is a design choice, not an SRS mandate.

Only authorised backend functions access C-DB. C-INT routes facts through domain rules; its worker handles integration records. Staff/external systems have no direct database access. Identity provisioning remains ASSUMP-02, without assumed registration screens.

The logical model below describes proposed information and relationships stored in C-DB, not a completed schema.

Entity / record	Main information and relationships	Responsible component / access	Integrity rule
Station	ID, location description, version; linked observations	C-MON reads; setup registers; C-INT routes observations	Completed moves reference registered stations. Missing facts remain unknown, never normal/zero.
PowerBank	ID, version, last confirmed station; separate location/charge/occupancy facts	C-DEV coordinates; domain modules query	Distinguish last-recorded from reliable current location; never invent a station.
FaultRecord	ID, device, description, unresolved/resolved condition, investigation progress, report/resolution actors/times/note, version	C-FAULT writes; C-MON/C-DEV/C-MOVE query	Registered device; ASSUMP-03 validation. Any unresolved fault excludes rental availability and redistribution.
RedistributionJob	ID, device, source/destination IDs, state, expected version, physical confirmation, actor/time	C-MOVE writes; C-DEV/C-MON read effects	Plans are not completed moves. Save completion/location atomically under Section 10.2 guards.
Observation / information quality	Entity/field/value, source, event/revision, observation/received times, quality/conflicting evidence	C-INT validates; C-DEV/C-MON accept/present	Preserve source evidence separately from accepted values. Arrival time alone cannot resolve conflicts. See Section 9.2.
Incoming processing record	Source/event ID, input fingerprint, result, affected record/version	C-INT and domain module	Source-scoped deduplication; reject conflicting ID reuse. Ordering depends on ASSUMP-05.
Command receipt	Identity/operation ID, input fingerprint, committed result/version	Domain module saves; C-AUTH checks access	Commit with business change. Identical authorised retries return saved results; failed saves leave no success receipt.
Pending outgoing update	Event/entity/version, payload, recipient, delivery state, attempts/acknowledgement	Domain transaction creates; C-INT delivers	Commit locally under DEC-04; remote acknowledgement is separate and does not prove hardware action.

Unique/foreign keys enforce identities and registered-device/station relationships; required/allowed-value constraints reject malformed records. Cross-record suitability rules remain backend transaction checks. See PostgreSQL constraints.

Separate operational facts from information quality

Store fault state, location, charge, occupancy and quality separately. A resolved fault proves neither charge nor presence/eligibility; missing/conflicting eligibility cannot establish suitability. A partial inventory cannot yield a confirmed exact total. [DEC-02; FR-02/03/05–07; NFR-02; SRC-01 pp.7–9; GAP-03/04]

Reported/investigating faults remain unresolved until a valid resolution commits. Progress names need not match a combined device-status field.

Keep observation and received times separate. Label useful previous evidence last-known; timestamps alone prove neither reliability nor a freshness guarantee.

Changes that must be saved together

- **Record a fault:** save the validated fault, increment the coordinated device version, and save the command receipt and any required outgoing update in one transaction.
- **Resolve a fault:** save the resolution and related version/receipt/outgoing changes together. Recheck remaining unresolved faults before deriving the device's exclusion result.
- **Complete redistribution:** recheck permission, suitability, expected device version and physical confirmation; save movement completion and the destination/version change together with the receipt and any outgoing update.
- **Accept an external event:** save its processing result with any accepted domain changes. Repeated input must not apply the same change again; contradictory evidence is retained for investigation rather than silently replacing a reliable value.

Return success only after commit; failure rolls back all related database changes (PostgreSQL transactions). Physical movement cannot roll back: investigate recording failure under Sections 9.3/10.2. Deliver external updates separately after commit. These are planned behaviours, not executed tests.

9.2 Caching and component data access

Cache scope and refresh rules

Under DEC-05, operational reads use C-DB without a separate business cache, reducing stale copies at the cost of database availability. Stored observations remain source/time-qualified evidence, not proof of a currently connected station or unchanged physical state.

C-UI keeps displayed results in memory, with no persistent offline operational store or write queue. Static assets may be cached without staff records; protected responses avoid persistent browser/shared caching.

Fetch fresh results on opening/manual refresh. Re-fetch affected device/fault/station results after successful changes, version conflicts or reconnection. Sign-out and denied/expired access clear protected data. Failed retrieval qualifies retained evidence as last-known; the Section 8 station view hides current counts while unavailable. Automatic polling/push and timing remain GAP-01.

Component requests and consistency

C-UI sends protected requests through C-AUTH. C-MON combines station observations with C-DEV/C-FAULT queries; device searches go to C-DEV. Fault and movement writes go through C-FAULT/C-MOVE, coordinated with C-DEV and C-DB under Section 9.1. C-INT routes validated incoming facts through these same domain functions and records outgoing delivery separately. Results include relevant quality, source/time and version information.

Routes are calls within the modular backend; restricted service credentials mediate C-DB access.

Assemble multi-query views in short read-only repeatable-read transactions, keeping fault/eligibility facts on one database snapshot (PostgreSQL isolation). This guarantees stored-data consistency, not physical freshness. Writes recheck current state under Section 10.

C-AUTH checks every protected request, including receipt retrieval; UI visibility never grants access or proves suitability. DEC-03; [OWASP authorisation]

Unavailable information and uncertain outcomes

Distinguish unknown, unavailable and conflicting facts under NFR-02. Preserve useful qualified evidence, never substitute zero/healthy/suitable. Database read failure shows unavailable; confirmed rollback shows save failed; only commit establishes a saved result.

A lost post-submission response leaves the outcome unconfirmed. Recheck/retry with the same operation ID/input, checking permission and receipt first. Local faults remain effective during outgoing-delivery failure; show synchronisation pending separately. Screens and Section 12 specify these paths, not execution results.

9.3 Backup, recovery and retention

Proposed backup approach

Propose scheduled PostgreSQL base backups plus continuous write-ahead-log (WAL) archiving. Restore a base backup and replay its uninterrupted log sequence to a consistent point; changes beyond recoverable material may be lost. See PostgreSQL recovery.

Recover the entire C-DB together: operational records, observations, receipts, incoming-processing and outgoing-update records. Preserve deployment configuration, database access definitions and recovery keys as well as data.

Encrypt recovery sets in restricted storage separate from the application/database host; control key access. Record backup/archive results and latest recoverable point. Never replace a usable set with an incomplete one. Agree schedules, storage and responsibilities under Section 9.4; no existing backups are claimed.

Recovery sequence and verification

1. **Contain:** suspend writes/outgoing delivery, show maintenance/unavailability and preserve incident evidence.

2. **Restore separately:** use an isolated environment, compatible versions and verified recovery material. Record the recovery point, errors and possible data-loss interval; keep delivery disabled.

3. **Verify:** check schema/access, IDs/relationships, unresolved-fault exclusion, movement/location/version agreement and matching receipts/outgoing records.

4. **Reconcile:** compare restored facts with station/Customer evidence and staff investigation. Qualify uncertain locations and exclude devices from confirmed suitability until resolved.

5. **Resume:** reopen only after checks/reconciliation succeed. Enable duplicate-safe processing/delivery and retain pending/investigation outcomes. Record the recovery decision and limitations.

The isolated restore exercise covers usable/corrupt recovery sets, restored exclusion, movement/location consistency, old observations, duplicates and uncertain delivery. Failed restore/checks block reopening. Measure restoration time and unrecovered interval against agreed targets; no executed exercise is claimed. [T-25]

Receipts created after the recovery point may be absent. Their absence cannot prove an action never happened: reconcile uncertain commands/delivery before replay. External deduplication/acknowledgement remains GAP-01/ASSUMP-05. Recovery cannot reverse physical movement or guarantee exactly-once external effects.

Retention and unconfirmed targets

GAP-08 leaves backup frequency, retention, acceptable loss and restoration time open; establish and validate them before deployment.

Agree retention for records, evidence, receipts and recovery sets. Preserve unresolved exclusion/relationships and deduplication through the supported retry period; older commands require review. Coordinate backup expiry to avoid silently reintroducing purged data on restore. No automatic purge interval is assumed.

9.4 Maintenance, updates and deployment

System part	Update/deployment approach	Compatibility and failure handling	Rollback/recovery
Operator interface	Build as a versioned web bundle. Staff get the new version on their next page load.	The backend accepts the current and previous interface version for one release. An out-of-date page that sends an incompatible request is rejected with a “refresh and retry” message and nothing is partially saved.	Redeploy the previous bundle. No data migration is needed.
Backend/business logic	Deploy a new version while the old one finishes in-flight requests, so a fault save is not dropped mid-write. Permission checks and version rechecks (DEC-03) are part of every release.	On startup the backend checks that the database schema version matches, and takes no traffic if it doesn’t. Changes commit fully or not at all, so a failed save is never shown as successful (FR-04, FR-06).	Redeploy the previous build, but only if the schema is still compatible.
Database/schema	Changes go through versioned, scripted migrations. We add new structures first, switch the code over, and only remove the old ones in a later release. A backup is taken before every migration.	We rehearse each migration on a copy loaded with synthetic test data (see Section 12). Existing faults, resolutions, movements and observations are kept, including each station’s last-response time.	If the change only added things, roll back the code and leave the data alone. If it removed or changed things, restore the pre-migration backup and manually review anything entered after it was taken.
Station/customer adapters	Each adapter is versioned against its agreed message format and deployed separately from the core logic, so a change on the station or Customer side doesn’t force a full release.	Messages we can’t recognise or that are corrupted are kept and flagged, not guessed at. Outgoing updates (DEC-04) wait in a queue while the other system is down and retry once it’s back.	Revert to the previous adapter version. Before replaying any queued messages, check for duplicates (ASSUMP-05).
External physical devices	These sit outside Ops (SRS Figure 1). Updating them is up to the station hardware supplier or ChargeMate. Who owns the station firmware isn’t established (GAP-01), so we don’t assume it, and Ops doesn’t install or update it.	If the station side changes its data format or behaviour, we treat it as an interface change (GAP-01, GAP-07).	Ops can roll back its own adapter. Until data arrives again, affected stations show as Unavailable with the time they last responded (NFR-02, SD-03).

10. Trade-offs and concurrent processes

10.1 Important trade-offs

These Section 3 choices balance reliability, usability and implementation cost; they are design decisions, not additional inherited requirements.

- **Consistency versus independence:** DEC-01 keeps related writes atomic in one modular backend/database, trading separate service scaling/releases for simpler coordination.
- **Freshness versus responsiveness:** DEC-02/05 expose qualified evidence and avoid another business cache. Reads depend on the database and screens may become stale; investigate uncertainty rather than display guessed zero/normal values. Add caching only for measured need.
- **Disconnected convenience versus valid writes:** DEC-03/05 require online, per-action permission/version checks instead of offline replay. Denial must be understandable; physical work during outages may need reconciliation. Permission assignments remain GAP-04.
- **Local durability versus immediate synchronisation:** DEC-04 commits the outgoing record with the change and delivers separately. Local success proves neither remote blocking nor exactly-once delivery; retain pending/uncertain status.
- **Recovery depth versus maintenance cost:** base backups plus archived changes support a later recovery point than periodic copies alone, but need usable logs, monitoring and restore checks. Numerical recovery targets remain GAP-08.

The design addresses qualitative usability, reliability and access without inventing update intervals/availability percentages. Recorded issues stay in scope; unconfirmed alert thresholds do not justify extra infrastructure. [SRC-01 pp.7–9; GAP-01/06/08]

10.2 Concurrent activities and race conditions

Overlapping operations can produce timing-dependent outcomes. Every fault, movement and accepted operational-fact change—including C-INT input—coordinates through C-DEV's device/version and a short transaction.

Concurrent events	Possible incorrect outcome	Coordination / guard	Expected result or recovery	Required Section 12 coverage and related specifications
Fault report versus movement confirmation	A stale selection admits a now-faulty device.	Lock device; recheck version, unresolved faults, suitability and physical confirmation.	Fault-first rejects stale movement. Move-first is not undone by a later fault; investigate conflicting physical evidence.	Both commit orders; record s/exclusion/messages. T-20
Two destination confirmations	Silent location overwrite.	Check device/source version and movement state/version in the same transaction.	Only compatible completion commits; conflict refreshes state. Reconcile any physical move already performed.	Competing destinations, same starting version. T-21
Resolution overlaps another fault	Resolution hides remaining fault/exclusion.	Same device lock/version; derive exclusion from all unresolved records.	Preserve remaining faults; stale versions require refresh.	T-13 covers multiple-fault baseline; concurrent-create extension remains unspecified.
Duplicate/late external events	Double application or stale overwrite.	Source/event ID, fingerprint and supported ordering (ASSUMP-05); same device transaction.	Return identical duplicate result; reject altered ID reuse; investigate incomparable contradictions.	Duplicates, older revisions, unordered conflicts. T-22
Lost response and retry	Duplicate fault/movement.	Check access and identity/operation ID/input receipt before treating an old version as a new command.	Return committed result; reject altered reuse. Missing restored receipts require reconciliation.	Lost response; identical/changed retries. T-23
Delivery/acknowledgement fails after local commit	Lost local fault or false remote success.	Atomic local change/outgoing record; stable event ID for retries.	Retain local commit and pending/uncertain delivery; no exactly-once external guarantee.	Delivery failure and lost acknowledgement. T-24

These are planned verification links, not execution results. The concurrent-create extension remains an additional specification obligation.

Coordination and retry rules

Writes use short read-committed transactions: reserve/check request/event ID, lock device, read current facts, validate version/rules and commit Section 9.1 changes. For multiple devices, lock by consistent device-ID order, then associated records consistently. Section 9.2's read-only repeatable-read snapshots do not replace write checks.

PostgreSQL locking supports conflicting row locks and consistent acquisition/deadlock retry. A lock cannot prioritise an unreceived fault or reverse physical action. Once a fault commits, later movement validation must exclude that device; a loaded screen cannot authorise completion.

Release database locks before staff movement/repair or external network waits. Preserve conflicting physical movement as investigation evidence with source/actor/time/destination; do not overwrite accepted location or report normal success. If evidence saving also fails, disclose that failure for later reconciliation. Existing location remains last-known.

Bounded transient rollback retries repeat all checks. Stale versions/uncertain physical facts need review. Changed input gets a new operation ID; unchanged retransmission keeps its ID. Retry/escalation limits remain GAP-08. Different devices can proceed concurrently, but device locks alone cannot enforce a future station-capacity rule (GAP-03).

11. Requirements Traceability Matrix

The five-column matrix covers all 11 Ops requirements. See Section 1.4 for summaries, Section 1.5 for use cases and Section 4.2 for component IDs. [SRC-01 pp.7–9; SRC-04]

Impact links identify implementation/interface/test dependencies for review, not stakeholder-approved changes to every linked SRS sentence. Links need not be symmetrical.

Test-link status: The references below link to the 26 detailed specifications in Section 12.3. They identify planned verification, not executed tests or passing results. Conditional acceptance rules remain documented in Section 12.4.

Requirement-ID from SRS	Other requirement IDs impacted if it changes	Related use cases	Responsible/affected system components	Related test cases
FR-01	FR-02, FR-08, NFR-01, NFR-02	UC-01 — View Station Status	C-UI, C-AUTH, C-MON, C-DEV, C-FAULT, C-INT, C-DB	T-01, T-02, T-03, T-17
FR-02	FR-01, FR-05, FR-07, FR-08, NFR-02	UC-01; availability checks in supporting redistribution scenario	C-UI, C-AUTH, C-MON, C-DEV, C-FAULT, C-INT, C-DB	T-04, T-05
FR-03	FR-04, FR-06, FR-07, FR-08, NFR-02	UC-02 — Track a Power Bank; device identification in UC-03	C-UI, C-AUTH, C-DEV, C-FAULT, C-INT, C-DB	T-06, T-07, T-08
FR-04	FR-02, FR-03, FR-05, FR-06, FR-07, FR-08	UC-03 — Report and Manage a Faulty Power Bank	C-UI, C-AUTH, C-DEV, C-FAULT, C-INT, C-DB	T-09, T-10, T-11, T-23
FR-05	FR-02, FR-03, FR-04, FR-06, FR-07, FR-08	UC-03; faulty-device information in UC-01/UC-02; supporting redistribution scenario	C-UI, C-AUTH, C-MON, C-DEV, C-FAULT, C-MOVE, C-INT, C-DB	T-05, T-09, T-13, T-16, T-20
FR-06	FR-02, FR-03, FR-04, FR-05, FR-07, FR-08	UC-03 — resolution branch	C-UI, C-AUTH, C-DEV, C-FAULT, C-INT, C-DB	T-12, T-13, T-14
FR-07	FR-01, FR-02, FR-03, FR-05, FR-08, NFR-02	Supporting redistribution scenario in Section 1.5; resulting station/device information in UC-01/UC-02	C-UI, C-AUTH, C-MON, C-DEV, C-FAULT, C-MOVE, C-INT, C-DB	T-15, T-16, T-20, T-21
FR-08	FR-01, FR-02, FR-03, FR-05, NFR-01, NFR-02	UC-01/UC-02/UC-03 alternative and failure branches; supporting issue-view scenario	C-UI, C-AUTH, C-MON, C-DEV, C-FAULT, C-MOVE, C-INT, C-DB	T-03, T-17
NFR-01	FR-01–FR-08	UC-01, UC-02, UC-03; supporting redistribution and issue-view scenarios	C-UI; C-MON, C-DEV, C-FAULT and C-MOVE provide task results and validation feedback	T-19
NFR-02	FR-01, FR-02, FR-03, FR-05, FR-07, FR-08	UC-01/UC-02; information used in UC-03 and supporting scenarios	C-UI, C-MON, C-DEV, C-MOVE, C-INT, C-DB	T-02, T-03, T-04, T-06, T-08, T-22, T-25
NFR-03	FR-01–FR-08, NFR-01	Every protected operational use case/scenario	C-AUTH; C-UI, C-MON, C-DEV, C-FAULT, C-MOVE, C-INT and C-DB apply the access boundaries	T-07, T-14, T-18, T-23

Fault rules affect availability/device/movement views; device identity affects correct association, and completed movement changes station views. C-AUTH/C-DB provide shared protection/storage without owning domain rules. DEC safeguards are additional design obligations, not new inherited requirements; redistribution elaborates FR-07 without asserting an approved UC-04. Customer ownership and conditional acceptance remain in Sections 3/12.4. Check all links in both directions.

12. Test specifications and test plan

12.1 Test approach and fixtures

Testing is proposed for a future implementation of the design in Sections 4 and 9–10. Expected behaviour is derived from the eight functional and three non-functional requirements and their fit criteria in SRC-01, pp.7–9. Design safeguards such as record versions, duplicate handling and recovery are additionally traced to the relevant DEC/ASSUMP entries. This section defines the approach and synthetic setup; it does not report completed tests or pass results.

Test levels and observations

Level	Focus	Evidence to collect during implementation testing
Component/domain	C-MON station summaries, C-DEV lookup, C-FAULT validation/exclusion, C-MOVE suitability and C-AUTH permission decisions, with controlled dependencies	Actual versus expected outputs for valid, missing, unknown, denied and conflicting inputs. Check that unresolved faults remain effective through reporting/investigation.
Integration	Domain modules with an isolated PostgreSQL database and C-INT connected to station/Customer simulators	Saved records, versions, command receipts and pending updates before/after each operation; transaction rollback, duplicates, delayed events and delivery failures.
End-to-end/operator	The proposed C-UI workflows for UC-01, UC-02, UC-03 and the supporting redistribution scenario	Visible navigation, labels, messages and resulting saved state. Representative staff perform normal tasks through the interface without database/command-line access or administrative help, as required by NFR-01.
Recovery/concurrency	The isolated restore process and overlapping operations described in Sections 9.3/10.2	Consistent restored records, qualified old information, conflict responses and duplicate protection. Control the order of competing submissions so both relevant commit orders can be examined.

The test harness may inspect stored records to verify outcomes; ordinary operator tasks must still use the normal interface. A failure message is insufficient evidence of rollback on its own: check that the relevant fault, movement, device version and receipt/outgoing records agree with the expected transaction outcome.

Coverage and case selection

The detailed specifications must collectively cover every inherited requirement, the main and alternative/failure branches of the three supplied use cases, and the FR-07 supporting redistribution scenario. Include registered versus unknown identifiers, required values present versus missing, permitted versus denied actions, confirmed zero availability versus unknown information, unresolved faults, valid/failed resolution and confirmed versus unconfirmed movement.

For the final state diagrams in Sections 6–8, identify tests for the permitted transitions, guarded rejections and relevant same-state queries. A rejected operation must leave the protected business state unchanged; a rejected or rolled-back new fault submission must not create a fault record. Resolving one fault must not clear another unresolved fault or independently establish rental eligibility. The concurrency scenarios in Section 10.2 also need explicit detailed cases.

Each case in Section 12.3 must identify its requirement/decision, chosen fixture variant, exact input, steps and expected UI/API and persisted outcomes. Section 12.2 supplies the case IDs; fixture IDs below are not test-case IDs. Record conditional or uncovered behaviour explicitly so the RTM does not imply coverage that has not been specified.

State-transition coverage

The detailed cases in Section 12.3 replace the repeated transition checklists. They specify future verification, not executed results. Use the independent fixture variants below; an error must not change protected business state.

- **FaultRecord (Section 6):** T-09, T-10, T-11, T-12, T-13, T-14, T-18, T-23, using DATA-01/02/04/06. Cover valid creation, rejected/rolled-back creation, investigation, successful/failed/denied resolution, multiple faults and lost-response retries. Retain resolved records; a remaining fault preserves exclusion, while resolution alone proves neither charge, presence nor rental eligibility. A lost response is unconfirmed, not proof of rollback. The Section 6 model uses these retained-history and eligibility rules. [SRC-01 p.19]

- **PowerBank tracking (Section 7):** T-06, T-07, T-08, using DATA-01/02/03. Cover unknown→known, conflicting→reconciled and rejected/unavailable searches. Preserve both unordered reports and keep unresolved conflicts flagged. Opening an unknown view must not erase accepted location. Closing/reopening ends only the lookup, without deleting the device, location/version or faults; location updates do not certify eligibility.
- **OperationalSnapshotQuality (Section 8):** T-01, T-02, T-03, T-04, T-17, T-26, using DATA-03/08. Cover initial Unknown, agreeing retrieval/refresh, unavailable retrieval, retry with agreeing/disagreeing/failed results, conflict/reconciliation and first-read conflicts. Unknown/unavailable is never confirmed zero; conflicting counts stay unconfirmed. Failures hide current counts without inventing a response time and preserve conflict evidence. Check all model starting states, including failure from Inconsistent; do not infer unlisted variants are specified. The table includes the first-read conflict path; the supplied figure does not show it explicitly. Archival from each active state is conditional on setup/permissions, preserves history and denies changes when unauthorised.

Across these checks, a reliable report that a station is offline remains reliable information; data quality and physical health are separate. Advancing time alone creates no freshness deadline or observation. Fixtures imply no new registration UI. Where case variants do not explicitly cover a model branch, record the gap rather than claiming coverage. [DEC-02; ASSUMP-02/04/05; GAP-01/03/04]

Synthetic fixture definitions

All identities, devices, faults and messages below are proposed test data, not production records, actual accounts or historical team activity. Reset the database and simulators before each independent case and apply only the stated variant. Complete-inventory, unavailable-source and conflicting-location variants are separate setups; do not combine them while retaining an unsupported confirmed total.

Fixture ID	Synthetic input / setup	Purpose	Relevant assumption / dependency
DATA-01	Two test identities, `USR-OPS-A` and `USR-OPS-B`, permitted for the tested action; `USR-VIEW` permitted to view but denied changes; `USR-NONE` unauthorised. Add an expired-session variant.	Allowed/denied requests, concurrent operators and changed access. These are test permission profiles, not proposed production staff roles.	ASSUMP-02; GAP-04; NFR-03
DATA-02	Registered `ST-01`/`ST-02`; baseline `PB-001` at ST-01, version 7, with no faults; `PB-002` at ST-01, version 3, with unresolved faults. For the complete-inventory variant, ST-02 has confirmed zero devices. `PB-999` is unregistered.	Station/device matching, known-empty versus unknown information, correct device association and movement setup.	ASSUMP-02; completeness/eligibility semantics remain GAP-03
DATA-03	Separate variants add registered `PB-003` with unknown location, registered `PB-004` reported at ST-01 by `SRC-A` and ST-02 by `SRC-B` without comparable ordering, or registered `ST-03` whose source cannot supply information. Retain observation/received times; include reliable-update, verified-reconciliation and still-conflicting variants.	Unknown/known/conflicting transitions, reconciliation and preservation of useful last-known evidence.	GAP-01; ASSUMP-05; NFR-02
DATA-04	`F-001` (“damaged connector”) and `F-002` (“charging issue”) both unresolved on PB-002. Use reported/investigation progress variants, plus a last-fault variant with F-002 already resolved and charge eligibility unknown. Fault-input variants use a non-empty/blank description; resolution variants use a non-empty/blank note and permitted/denied identities.	Correct fault creation, missing details, multiple-fault exclusion and successful/failed/denied resolution.	ASSUMP-03/04; final field/permission rules remain GAP-04
DATA-05	Proposed `MOVE-001`: PB-001, ST-01 → ST-02, expected version 7, physical confirmation true/false. For competing destinations, add registered available ST-04 and submit ST-02 versus ST-04 confirmations against the same starting version.	Confirmed movement, missing confirmation, outdated versions and competing reports.	Charge/capacity inputs use a configured test policy; actual fault, registration, version and confirmation guards still run. GAP-03/05
DATA-06	Source `SRC-STATION` sends event `EVT-001`, revision 10; replay identical input, conflicting input with the same ID, and a different event with older revision 9. Command variants reuse `OP-001` unchanged or with altered input; outgoing variants lose delivery/acknowledgement.	Duplicate/late messages, lost responses, command receipts and pending synchronisation.	ASSUMP-05; acknowledgement/recipient rules remain GAP-01

Fixture ID	Synthetic input / setup	Purpose	Relevant assumption / dependency
DATA-07	Prepare a consistent recovery set containing unresolved faults, a completed movement with matching location/version, receipts and pending outgoing updates. Add an unusable recovery-set variant, such as a missing required archived log.	Restore consistency, unrecovered intervals, old observations and uncertain replay. These are setup definitions, not existing backup files.	Section 9.3; recovery/retention targets remain GAP-08
DATA-08	Registered test snapshots: `ST-Q01` starts Unknown with no facts; `ST-Q02` starts Reliable with an accepted count of 6 and a recorded response time. Variants provide count 0 from a complete inventory, comparable source counts 6/5, a reliable offline-station report, failed retrieval, recovery and permitted/denied archival.	Station-quality transitions, zero versus unknown, preserved conflict evidence and quality versus health.	Section 8; ASSUMP-02; external field/completeness and archival permissions remain GAP-01/03/04

Controlled failures and unresolved acceptance rules

Station/Customer simulators use the proposed Section 4.4 fields to produce successful, missing, conflicting, duplicate, delayed and unavailable responses without operating real rental hardware. Inject database failure before commit, loss of the response after commit and external delivery/acknowledgement failure as distinct conditions. A confirmed rollback leaves no successful business update; a lost post-submission response leaves the outcome unconfirmed until checked using the original operation ID. A local commit can coexist with pending external delivery.

Movement fixtures may supply controlled charge/capacity facts to exercise transaction mechanics while the actual fault, registration, version and confirmation rules remain active. This does not validate an unconfirmed charge threshold or capacity policy. Numerical freshness, charge, alert, performance, recovery and retention limits are not invented for testing. Cases depending on those values, exact exchanged fields or permission assignments must identify their gap and conditional expected result; the missing rule must be established before the corresponding acceptance claim can be finalised.

12.2 Test inventory

Test ID	Description	Requirement/decision	Main/success/failure branch	Detailed specification location
T-01	View station status with reliable information	FR-01, FR-02, NFR-01, NFR-02	Main success (UC-01)	Section 12.3, T-01
T-02	Station information unavailable, then retry	FR-01, NFR-02; SD-03	Alternative: unavailable; retry success/failure	Section 12.3, T-02
T-03	Conflicting station information flagged and cleared	FR-01, FR-08, NFR-02; SD-03	Alternative: inconsistent; reconciliation	Section 12.3, T-03
T-04	Confirmed zero availability versus unknown availability	FR-02, NFR-02	Success and failure distinction	Section 12.3, T-04
T-05	Faulty device excluded from normal availability	FR-02, FR-05, DEC-02	Main success	Section 12.3, T-05
T-06	Track a device: unknown, then known location	FR-03, NFR-02; SD-02	Main success (UC-02)	Section 12.3, T-06
T-07	Track a device: unknown ID, retrieval failure, denied access	FR-03, NFR-03	Failure (UC-02)	Section 12.3, T-07
T-08	Conflicting device location and reconciliation	FR-03, NFR-02; SD-02	Alternative: conflicting	Section 12.3, T-08
T-09	Record a valid fault	FR-04, FR-05; SD-01	Main success (UC-03)	Section 12.3, T-09
T-10	Fault rejected for missing details	FR-04	Failure: validation	Section 12.3, T-10
T-11	Fault save fails and rolls back	FR-04, NFR-02	Failure: failed save	Section 12.3, T-11
T-12	Review and successfully resolve a fault	FR-06, FR-05; SD-01; DEC-02	Main success: resolution	Section 12.3, T-12
T-13	Resolve one of several faults; exclusion persists	FR-05, FR-06; ASSUMP-04	Alternative: multiple faults	Section 12.3, T-13
T-14	Resolution rejected: blank note, no permission, failed save	FR-06, NFR-03; SD-01	Failure	Section 12.3, T-14
T-15	Confirmed redistribution of a suitable device	FR-07	Main success	Section 12.3, T-15
T-16	Redistribution rejected: faulty device, no confirmation, invalid destination	FR-07, FR-05	Failure	Section 12.3, T-16
T-17	Recorded issues distinguished from normal operation	FR-08; DEC-06	Main success	Section 12.3, T-17
T-18	Access control on direct backend requests	NFR-03; DEC-03	Success and failure	Section 12.3, T-18
T-19	Operator tasks via visible controls only	NFR-01	Usability	Section 12.3, T-19
T-20	Fault report versus movement confirmation race	DEC-03; Section 10.2	Concurrency (both orders)	Section 12.3, T-20
T-21	Two operators confirm different destinations	DEC-03; Section 10.2	Concurrency	Section 12.3, T-21
T-22	Duplicate, late and conflicting external events	NFR-02; ASSUMP-05	Integration	Section 12.3, T-22
T-23	Lost response and retry with the same operation ID	FR-04; DEC-03; Section 10.2	Failure/recovery	Section 12.3, T-23
T-24	Outgoing delivery fails after local commit	DEC-04; NFR-02	Failure/recovery	Section 12.3, T-24
T-25	Isolated restore and consistency check	Section 9.3; NFR-02	Success and failure	Section 12.3, T-25
T-26	Station snapshot archival, permitted and denied	SD-03; GAP-04	Conditional	Section 12.3, T-26

12.3 Detailed test-case specifications

T-01 — View station status with reliable information

Field	Specification
Requirement(s) / design decision	FR-01, FR-02, NFR-01, NFR-02
Description and objective	Check an operator can open monitoring and see a selected station's latest available information with source and time.
Preconditions / initial state	USR-OPS-A signed in. ST-Q02 is Reliable (count 6, recorded response time) from DATA-08.
Input specification / exact test data	DATA-01 (USR-OPS-A), DATA-08 (ST-Q02).
Steps	1. Open the station overview. 2. Select ST-Q02.
Expected output specification	Overview lists ST-Q02. Detail shows status, availability 6 and a "Last updated" time with source.
Pass criteria	Displayed values equal the stored values. No Unknown or attention label appears.
Cleanup	Standard reset
Dependencies / unresolved rules	Exact exchanged fields: GAP-01.

T-02 — Station information unavailable, then retry

Field	Specification
Requirement(s) / design decision	FR-01, NFR-02; SD-03 (Unknown/Reliable → Unavailable → Reliable)
Description and objective	Check failed retrieval shows Unavailable instead of stale or zero values, and that retry outcomes are handled correctly.
Preconditions / initial state	Variant A starts ST-Q01 (Unknown). Variant B starts ST-Q02 (Reliable, count 6).
Input specification / exact test data	DATA-08, DATA-03 (ST-03 source unavailable). Simulator returns a failure. Retry variants return agreeing data (C) or fail again (D).
Steps	1. Trigger retrieval with the simulator failing. 2. View the station. 3. Retry with agreeing data (C), or with a failure (D).
Expected output specification	A/B: "Unavailable" and "—", no number; last-response time kept for B. C: Reliable, Unavailable label removed. D: stays Unavailable with an error.
Pass criteria	No count of 0 or 6 shown while Unavailable. A failed attempt does not update the response time.
Cleanup	Standard reset
Dependencies / unresolved rules	No freshness deadline is used (GAP-01).

T-03 — Conflicting station information flagged and cleared

Field	Specification
Requirement(s) / design decision	FR-01, FR-08, NFR-02; SD-03
Description and objective	Check disagreeing sources are both kept and flagged, and that the flag clears only when sources agree.
Preconditions / initial state	ST-Q02 Reliable.
Input specification / exact test data	DATA-08 comparable counts 6 (Rental Station System) and 5 (Customer records). Variant C: all sources then report 6. Variant D: first retrieval for ST-Q01 already returns 6/5. Variant E: reliable offline-station report.
Steps	1. Supply the 6/5 reports. 2. View the station. 3. Supply agreeing data (C). 4. Run D and E separately.
Expected output specification	Both values and sources shown with "Needs investigation". After C, the agreed value shows and the flag is removed. D shows the flag. E stays Reliable.
Pass criteria	Neither value is shown as confirmed while they disagree. A reliable offline report is not treated as poor-quality data.
Cleanup	Standard reset
Dependencies / unresolved rules	Variant D is explicit in the Section 8 table but not the supplied figure; model alignment remains a review condition.

T-04 — Confirmed zero availability versus unknown availability

Field	Specification
Requirement(s) / design decision	FR-02, NFR-02
Description and objective	Check a known-empty station and a station with no data look different.
Preconditions / initial state	Variant A: ST-02 complete inventory, zero devices. Variant B: ST-Q01 Unknown. Variant C: partial device list.
Input specification / exact test data	DATA-02 complete-inventory variant, DATA-08 ST-Q01.
Steps	1. View ST-02. 2. View ST-Q01. 3. View the partial-list station.
Expected output specification	A shows 0 as confirmed. B shows "Unknown" and "—". C shows no exact total.
Pass criteria	0 appears only in A.
Cleanup	Standard reset
Dependencies / unresolved rules	What counts as a complete inventory: GAP-03.

T-05 — Faulty device excluded from normal availability

Field	Specification
Requirement(s) / design decision	FR-02, FR-05, DEC-02
Description and objective	Check unresolved faults visibly exclude a device from normal availability.
Preconditions / initial state	PB-001 (no faults) and PB-002 (F-001, F-002 unresolved) both at ST-01.
Input specification / exact test data	DATA-02, DATA-04.
Steps	1. Open ST-01 availability. 2. Open the PB-002 device view.
Expected output specification	PB-001 counted as available. PB-002 not counted and marked as requiring attention.
Pass criteria	The exclusion comes from the unresolved faults, not a separate flag.
Cleanup	Standard reset
Dependencies / unresolved rules	Charge eligibility: GAP-03.

T-06 — Track a device: unknown, then known location

Field	Specification
Requirement(s) / design decision	FR-03, NFR-02; SD-02 (Initial → Unknown → Known → Final)
Description and objective	Check lookup shows unknown until a reliable observation arrives, then the location, and closes cleanly.
Preconditions / initial state	PB-003 registered, no reliable location.
Input specification / exact test data	DATA-01, DATA-03 (PB-003, reliable ST-01 observation with source and time).
Steps	1. Search PB-003. 2. Supply the reliable observation. 3. Review the result (resultReviewed()). 4. Search again.
Expected output specification	Step 1: "unknown", no invented station. Step 2: location ST-01. Step 3: lookup closed, device and fault records unchanged. Step 4: information retrieved again.
Pass criteria	Lookup never alters device version, faults or charge facts.
Cleanup	Standard reset
Dependencies / unresolved rules	Event names are provisional (Section 7).

T-07 — Track a device: unknown ID, retrieval failure, denied access

Field	Specification
Requirement(s) / design decision	FR-03, NFR-03
Description and objective	Check each failed search gives the right distinct outcome and stores nothing.
Preconditions / initial state	PB-001 registered.
Input specification / exact test data	PB-999 (unregistered) with USR-OPS-A. PB-001 with retrieval failure injected. PB-001 with USR-NONE.
Steps	1. Run the three variants separately.
Expected output specification	Not found. Unavailable/read failed. Access denied with no device data shown.
Pass criteria	No lookup opened, no stored record changed, and the three messages differ.
Cleanup	Standard reset
Dependencies / unresolved rules	None.

T-08 — Conflicting device location and reconciliation

Field	Specification
Requirement(s) / design decision	FR-03, NFR-02; SD-02; ASSUMP-05
Description and objective	Check disagreeing reports are preserved and the flag clears only on a reliable outcome.
Preconditions / initial state	PB-004 reported at ST-01 by SRC-A and ST-02 by SRC-B, no comparable ordering.
Input specification / exact test data	DATA-03 conflicting variant, then verified-reconciliation (ST-02) and still-conflicting variants.
Steps	1. Search PB-004. 2. Apply the verified reconciliation (variant A).3. Repeat with no reliable outcome (variant B).
Expected output specification	"Requires investigation" with both reports kept. After A, location ST-02 and the flag cleared. B stays conflicting.
Pass criteria	The last-arrived report is never chosen automatically.
Cleanup	Standard reset
Dependencies / unresolved rules	Who may reconcile: GAP-04.

T-09 — Record a valid fault

Field	Specification
Requirement(s) / design decision	FR-04, FR-05; SD-01 (Initial → Reported)
Description and objective	Check a valid fault is saved against the correct device and excludes it.
Preconditions / initial state	PB-001 at ST-01, version 7, no faults. USR-OPS-A permitted.
Input specification / exact test data	DATA-04 non-empty variant: device PB-001, description "cracked casing", operation ID OP-001.
Steps	1. Find PB-001. 2. Submit the fault.
Expected output specification	New fault linked to PB-001, state Reported. Device version 8. Receipt and pending outgoing update saved. Success shown only after commit. PB-001 shown as requiring attention and excluded.
Pass criteria	One fault stored against the correct device.
Cleanup	Standard reset
Dependencies / unresolved rules	Final field list: GAP-04.

T-10 — Fault rejected for missing details

Field	Specification
Requirement(s) / design decision	FR-04
Description and objective	Check an incomplete fault is rejected with a clear message.
Preconditions / initial state	As T-09.
Input specification / exact test data	DATA-04 blank-description variant for PB-001. A second variant has no device selected.
Steps	1. Submit each incomplete form.
Expected output specification	Validation rejection naming the missing item.
Pass criteria	No fault, receipt or outgoing record created. Device version stays 7.
Cleanup	Standard reset
Dependencies / unresolved rules	ASSUMP-03 validation rules.

T-11 — Fault save fails and rolls back

Field	Specification
Requirement(s) / design decision	FR-04, NFR-02
Description and objective	Check a failed save is never shown as successful.
Preconditions / initial state	As T-09.
Input specification / exact test data	Valid fault for PB-001, with a database failure injected before commit.
Steps	1. Submit the fault.
Expected output specification	"Save failed" message. Fault, receipt and outgoing update all absent. Version stays 7.
Pass criteria	No success message. No partial record.
Cleanup	Standard reset
Dependencies / unresolved rules	None.

T-12 — Review and successfully resolve a fault

Field	Specification
Requirement(s) / design decision	FR-06, FR-05; SD-01; DEC-02
Description and objective	Check an authorised resolution is stored, the record is kept, and resolution alone does not make the device rental-ready.
Preconditions / initial state	PB-001 has one unresolved fault F-003 in state Reported.
Input specification / exact test data	USR-OPS-A; resolution note "connector replaced".
Steps	1. Review F-003. 2. Submit the resolution.
Expected output specification	F-003 moves to Under investigation, then Resolved with note, actor and time. Record retained, not deleted. Device version incremented. Outgoing update saved.
Pass criteria	Exclusion clears only because no unresolved fault remains. No "rental ready" is shown, and charge and location are unchanged.
Cleanup	Standard reset
Dependencies / unresolved rules	Resolver permission: GAP-04.

T-13 — Resolve one of several faults; exclusion persists

Field	Specification
Requirement(s) / design decision	FR-05, FR-06; ASSUMP-04
Description and objective	Check clearing one fault does not hide another.
Preconditions / initial state	PB-002 with F-001 and F-002 unresolved.
Input specification / exact test data	Valid resolution of F-001, then of F-002.
Steps	1. Resolve F-001. 2. View PB-002 and ST-01 availability. 3. Resolve F-002. 4. View again.
Expected output specification	After step 1, PB-002 is still excluded and requiring attention. After step 3, exclusion clears.
Pass criteria	Exclusion follows the count of unresolved faults.
Cleanup	Standard reset
Dependencies / unresolved rules	ASSUMP-04 is unconfirmed.

T-14 — Resolution rejected: blank note, no permission, failed save

Field	Specification
Requirement(s) / design decision	FR-06, NFR-03; SD-01 (same-state outcomes)
Description and objective	Check each failed resolution leaves the fault unresolved.
Preconditions / initial state	F-003 Under investigation.
Input specification / exact test data	Variant A blank note. Variant B `USR-VIEW`. Variant C database failure before commit.
Steps	1. Run each variant separately.
Expected output specification	Validation error, access denied, "save failed" respectively.
Pass criteria	F-003 stays unresolved, device version unchanged, nothing shown as successful.
Cleanup	Standard reset
Dependencies / unresolved rules	GAP-04.

T-15 — Confirmed redistribution of a suitable device

Field	Specification
Requirement(s) / design decision	FR-07
Description and objective	Check a confirmed move records the new location.
Preconditions / initial state	PB-001 at ST-01, version 7, no faults.
Input specification / exact test data	DATA-05 MOVE-001 to ST-02, expected version 7, physical confirmation true. The test policy supplies charge and capacity.
Steps	1. Select the device and destination. 2. Confirm physical movement. 3. Complete.
Expected output specification	Movement completed, PB-001 at ST-02, version 8, receipt and outgoing update saved. Both station views updated.
Pass criteria	Location changes only after confirmation and commit.
Cleanup	Standard reset
Dependencies / unresolved rules	Charge and capacity rules: GAP-03, GAP-05.

T-16 — Redistribution rejected

Field	Specification
Requirement(s) / design decision	FR-07, FR-05
Description and objective	Check unsuitable moves are refused.
Preconditions / initial state	As T-15, plus PB-002.
Input specification / exact test data	Variant A PB-002 (unresolved faults). Variant B confirmation false. Variant C unregistered destination.
Steps	1. Attempt each variant.
Expected output specification	A: not offered or rejected. B: no completion. C: rejected.
Pass criteria	Locations and versions unchanged, with a clear reason shown.
Cleanup	Standard reset
Dependencies / unresolved rules	GAP-05.

T-17 — Recorded issues distinguished from normal operation

Field	Specification
Requirement(s) / design decision	FR-08; DEC-06
Description and objective	Check issues are visible and distinct.
Preconditions / initial state	PB-002 faulty, ST-03 unavailable, ST-Q02 6/5 conflict, one normal station.
Input specification / exact test data	DATA-02, DATA-03, DATA-08.
Steps	1. Open the attention view (WF-03).
Expected output specification	Each issue shown with its own type and reason. The normal station has no indicator.
Pass criteria	No automatic alert or threshold appears.
Cleanup	Standard reset
Dependencies / unresolved rules	Alert rules: GAP-06.

T-18 — Access control on direct backend requests

Field	Specification
Requirement(s) / design decision	NFR-03; DEC-03
Description and objective	Check protected actions are denied without relying on the UI.
Preconditions / initial state	DATA-01 profiles.
Input specification / exact test data	Direct requests to record a fault, resolve and confirm movement as USR-NONE, USR-VIEW, an expired session, and USR-OPS-A.
Steps	1. Send each request.
Expected output specification	Denied for the first three. The permitted request commits.
Pass criteria	Denied requests change no record and reveal no protected data.
Cleanup	Standard reset
Dependencies / unresolved rules	Permission matrix: GAP-04.

T-19 — Operator tasks via visible controls only

Field	Specification
Requirement(s) / design decision	NFR-01
Description and objective	Check ordinary tasks need no technical help.
Preconditions / initial state	UI built (WF-01 to WF-08).
Input specification / exact test data	Three task scripts: view a station, find a device, record a fault. Representative staff using USR-OPS-A.
Steps	1. Observer gives each script. 2. Participant completes it.
Expected output specification	Tasks completed through visible controls.
Pass criteria	No database access, command line or administrator help needed. Record any interventions.
Cleanup	Standard reset
Dependencies / unresolved rules	No numeric time target exists (GAP-08), so record times without a pass limit.

T-20 — Fault report versus movement confirmation race

Field	Specification
Requirement(s) / design decision	DEC-03; Section 10.2
Description and objective	Check the result is correct in both commit orders.
Preconditions / initial state	PB-001 version 7.
Input specification / exact test data	Fault by USR-OPS-A, MOVE-001 by USR-OPS-B, released together with a controlled barrier. Run order 1 (fault first) and order 2 (movement first).
Steps	1. Run each order. 2. Inspect stored records.
Expected output specification	Order 1: movement rejected or conflicted, location stays ST-01. Order 2: movement completes, then the fault saves against the device.
Pass criteria	No silent loss. The device is excluded once the fault is committed.
Cleanup	Standard reset
Dependencies / unresolved rules	GAP-05.

T-21 — Two operators confirm different destinations

Field	Specification
Requirement(s) / design decision	DEC-03; Section 10.2
Description and objective	Check competing reports cannot both commit.
Preconditions / initial state	PB-001 version 7.
Input specification / exact test data	ST-02 by USR-OPS-A and ST-04 by USR-OPS-B, same starting version.
Steps	1. Submit both at once.
Expected output specification	One commits (version 8). The other gets a conflict and refreshed data.
Pass criteria	The device has exactly one recorded location.
Cleanup	Standard reset
Dependencies / unresolved rules	GAP-05.

T-22 — Duplicate, late and conflicting external events

Field	Specification
Requirement(s) / design decision	NFR-02; ASSUMP-05
Description and objective	Check repeated or out-of-order input is handled safely.
Preconditions / initial state	SRC-STATION configured.
Input specification / exact test data	EVT-001 revision 10. Then an identical replay, the same ID with different input, and revision 9 after revision 10.
Steps	1. Send each message.
Expected output specification	Applied once. The replay returns the stored result. Conflicting reuse is rejected. Revision 9 does not overwrite revision 10.
Pass criteria	No double-apply, and incomparable evidence is retained.
Cleanup	Standard reset
Dependencies / unresolved rules	Ordering support: ASSUMP-05, GAP-01.

T-23 — Lost response and retry with the same operation ID

Field	Specification
Requirement(s) / design decision	FR-04; DEC-03; Section 10.2
Description and objective	Check a retry never creates a second change.
Preconditions / initial state	As T-09.
Input specification / exact test data	OP-001 fault submission with the response dropped after commit. Retry unchanged. Retry with altered input. Retry as USR-NONE.
Steps	1. Submit and drop the response. 2. Retry each variant.
Expected output specification	Unchanged retry returns the saved result. Altered input is rejected. USR-NONE is denied.
Pass criteria	Exactly one fault exists.
Cleanup	Standard reset
Dependencies / unresolved rules	None.

T-24 — Outgoing delivery fails after local commit

Field	Specification
Requirement(s) / design decision	DEC-04; NFR-02
Description and objective	Check a saved fault survives delivery failure and the status is honest.
Preconditions / initial state	Customer simulator unreachable.
Input specification / exact test data	Valid fault (as T-09). Then a lost-acknowledgement variant.
Steps	1. Save the fault. 2. View status. 3. Restore the simulator.
Expected output specification	Fault stays committed. "Saved locally, sync pending". Retry uses the same event ID. Lost acknowledgement stays "uncertain".
Pass criteria	The interface never claims remote blocking or confirmed delivery.
Cleanup	Standard reset
Dependencies / unresolved rules	Recipient handling: GAP-01, GAP-07.

T-25 — Isolated restore and consistency check

Field	Specification
Requirement(s) / design decision	Section 9.3; NFR-02
Description and objective	Check a restore produces consistent data and a bad restore blocks service.
Preconditions / initial state	Isolated environment.
Input specification / exact test data	DATA-07 usable set, then the unusable variant (missing archived log).
Steps	1. Restore. 2. Run consistency checks. 3. Repeat with the unusable set.
Expected output specification	Usable: unresolved faults still exclude, movements match locations and versions, old observations labelled last-known. Unusable: restore fails.
Pass criteria	Normal service does not resume after a failed restore or failed check. Record elapsed time and unrecovered interval.
Cleanup	Standard reset
Dependencies / unresolved rules	Targets: GAP-08.

T-26 — Station snapshot archival, permitted and denied

Field	Specification
Requirement(s) / design decision	SD-03 (Any state → Final); GAP-04
Description and objective	Check archival ends the snapshot lifecycle only when permitted.
Preconditions / initial state	Snapshots in Unknown, Reliable, Unavailable and Inconsistent states.
Input specification / exact test data	DATA-08 permitted and denied profiles.
Steps	1. Attempt archival from each state.
Expected output specification	Permitted: lifecycle ends, history kept. Denied: no change.
Pass criteria	No history lost, and denial changes nothing.
Cleanup	Standard reset
Dependencies / unresolved rules	Conditional on the permission policy (GAP-04, ASSUMP-02).

12.4 Test plan

Schedule and milestones

Milestone	Entry condition	Testing activities / test IDs	Resources and responsible role/person	Proposed timing/ dependency	Deliverables / exit condition
M0 — baseline ready	Sections 1–11 reviewed. DATA-01 to DATA-08 agreed.	Review T-01 to T-26 against the RTM and state diagrams. Prepare fixtures and simulators. No execution.	Test lead (Harrison, proposed), RTM reviewer (Yousef), screen checker (Cooper)	Before implementation starts	Reviewed inventory. Every requirement has a test. Blocked tests identified.
M1 — domain logic ready	C-AUTH, C-DEV, C-FAULT, C-MON and C-MOVE built against an isolated database.	T-04 to T-18 (backend level), T-26	Test lead with module developers	After M0 and each module's completion	Recorded results. Defects logged.
M2 — integration ready	C-INT and simulators connected to the domain modules.	T-02, T-03, T-17, T-20 to T-25	Test lead, integration developer, isolated restore environment	After M1 exit	Integration results. Race and recovery outcomes recorded.
M3 — UI / acceptance ready	C-UI built (WF-01 to WF-08).	T-01, T-19, plus visible-outcome re-runs of T-06 to T-16	Test lead, representative staff, Cooper for screen behaviour	After M2 exit	Acceptance results. Usability observations.
M4 — release candidate	All open defects triaged.	Full regression of T-01 to T-26 and any conditional tests whose gap has closed	Whole team	After M3 exit	Final results against the RTM. Remaining gaps listed.

Environments, resources and reporting

- **Browsers and devices:** a supported desktop browser and a tablet browser (ASSUMP-01).
- **Database:** an isolated PostgreSQL test instance (DEC-01), reset to fixtures between cases.
- **Simulators:** Rental Station System and Customer simulators using the Section 4.4 fields. They produce successful, missing, conflicting, duplicate, delayed and unavailable responses. No real hardware is operated.
- **Test data and users:** DATA-01 to DATA-08, plus representative staff for T-19.
- **Tools:** an automated test runner for the chosen backend language, and browser automation for UI flows. Specific tools are to be confirmed.
- **Defect recording:** one GitHub issue per defect with test ID, fixture, expected and actual result, and severity.
- **Retest and regression:** a fixed defect is retested with its own case, then the affected requirement's cases are re-run before M4.
- **Acceptance reporting:** each result records build/version, data, date, executor, actual result and evidence. The RTM is updated per requirement.

Blocked or conditional test specifications

Test ID	Missing rule / prerequisite	Gap ID	What can be specified now	What must be confirmed
T-15, T-16	Charge and capacity rules for suitability	GAP-03, GAP-05	Transaction mechanics using a test policy	Real suitability and capacity rules
T-14, T-18, T-26	Who may resolve faults, and archival permissions	GAP-04	Permitted versus denied behaviour with test profiles	Real permission assignments
T-22	Event ordering support and exchanged fields	GAP-01, ASSUMP-05	Duplicate and conflicting-ID handling	Revision and ordering contract
T-24	Recipient acknowledgement and remote blocking	GAP-01, GAP-07	Local commit, pending status, retry	Recipient handling
T-25, T-19	Recovery and usability targets	GAP-08	Consistency and failure behaviour, recorded times	Numerical targets
T-03 (variant D)	First-retrieval conflict absent from supplied figure	—	Expected result and Unknown → Inconsistent table row	Diagram/table alignment remains open

13. Report to COMP2050 staff

13.1 Reflection on using the received SRS

The Ops SRS was useful because its eight FRs, three NFRs and observable fit criteria gave architecture and tests common targets. FR-04, for example, requires a fault saved against the correct registered device and visible failure when saving fails. Those conditions drove the domain and transaction boundaries. [SRC-01, pp.7–9]

The use cases distinguish not-found, unknown and conflicting location outcomes. NFR-02 therefore requires more than success screens: Sections 4/9 preserve these distinctions in views, observations and failure handling. [SRC-01, pp.12–19]

The harder issue was Customer/Ops allocation. Customer expects availability/charge/capacity and release/return exchanges that Ops' FRs do not fully assign. We defined C-INT and GAP-01–GAP-03 rather than silently adding payment or hardware-control responsibilities. This supports planning while exposing the missing contract. [SRC-03, pp.7–12,15,23–24; SRC-01, pp.5,7–9,37]

The video describes blocking and restoring active inventory, but the written SRS says resolution alone establishes neither charge, presence nor rental eligibility. Section 9.1 therefore separates these facts; ASSUMP-04 preserves exclusion while another fault remains unresolved. The multiple-fault policy still needs confirmation. [SRC-02, 7:26–7:49; SRC-01, pp.18–19; DEC-02]

FR-07 requires confirmed movement, yet the SRS details only UC-01–UC-03. We elaborated a supporting scenario and save-time checks in Sections 9–10. Failed recording after physical movement remains a reconciliation problem: rollback cannot reverse the move. Capacity/exception rules stay open. [SRC-01, pp.8,11,19,32,36–38; GAP-03/05]

Document changes are evidenced by Section 3 draft, Section 4 draft, storage draft, concurrency draft, RTM draft. These commits establish revisions, not stakeholder approval, peer sign-off or executed tests.

Separating gaps, assumptions and decisions allowed progress without disguising provisional rules. PostgreSQL and module labels are justified design choices; charge thresholds, refresh deadlines, permission assignments and recovery targets require business agreement. This limits acceptance claims even when a technical mechanism can already be specified.

A stronger handover would add a field/status dictionary, interface ownership, complete redistribution exceptions and measurable operational targets. Consistent written/video claims about freshness, hardware control and eligibility would reduce interpretation across screens, models and tests.

The result is a traceable design with linked tests and explicit unresolved rules. Final review must align state wording and conditional acceptance; later stakeholder responses or team reviews need genuine evidence and dates.

13.2 Tools, communication and changes from A1 to A2

Activity	What the team actually used in A1	What the team actually used in A2	What changed and its effect	Evidence
Communication and clarification	Joint WhatsApp channels with Team B, plus six minuted meetings with all four members present	WhatsApp group chat	A1 had a stakeholder who could answer questions. In A2 the gaps in the SRS had to be handled as documented assumptions	A1 SRS Appendices A–B; Section 3 (GAP-01 to GAP-08)
Tasks and sprint planning	Tasks allocated in meetings and recorded in the minutes (e.g. UC-01 to Cooper, UC-02 to Harrison, UC-03 to Isaac)	Sections allocated by owner and reviewer in the SDD working guide	Each section had a named owner and reviewer, so it was clear who was responsible, but most of the integration still happened close to the deadline	A1 minutes, 18 August 2026; SDD section headers
Document and diagram authoring	SRS in Markdown on GitHub; diagrams in Draw.io and Lucid.app	SDD in Markdown on GitHub; state diagrams in Draw.io (SD-01, SD-02) and Lucidchart (SD-03); wireframes in Lucidchart (WF-01 to WF-08)	Design needed far more visual work than requirements did, so Lucidchart was added for drawing screens and diagrams	Section 5 and 6–8 figures; REF-06, REF-07
Review and version control	GitHub with pull requests (PR #1, PR #2) and a change log tracking who changed what	GitHub `a2docs` folder, with an owner and reviewer listed on each section	Reviewers were named for every section, so cross checking was planned instead of happening only at the end	A1 SRS change log; SDD section headers; repository history

The main improvement for future work is earlier integration: named ownership clarified responsibility, but late merging left cross-section checks close to submission. The repository history in Section 13.3 supports this limitation.

13.3 Sprint allocations and repository evidence

The brief requires three two-week sprints. Repository history supports the revisions below, and Appendix A contains the team-supplied WhatsApp-derived interaction records. Original sprint allocations and GitHub burn-up exports were not supplied; no sprint dates, assignments or completion totals are invented.

Required period	Evidence available	What can be concluded	Limitation
Sprint 1: two weeks from the end of Week 8	No dated sprint allocation or chart supplied.	No verified allocation/completion statement can be made.	Original snapshot and burn-up evidence missing.
Sprint 2: two weeks from the end of the mid-session break	No dated sprint allocation or chart supplied.	No verified allocation/completion statement can be made.	Original snapshot and burn-up evidence missing.
Sprint 3: two weeks from the end of Week 10	A2 document revisions dated 31 October–1 November 2026 are present in repository history.	Substantial document integration took place near the deadline.	Commits do not establish the original sprint allocation, estimates, task completion or review approval.

Recorded document progress

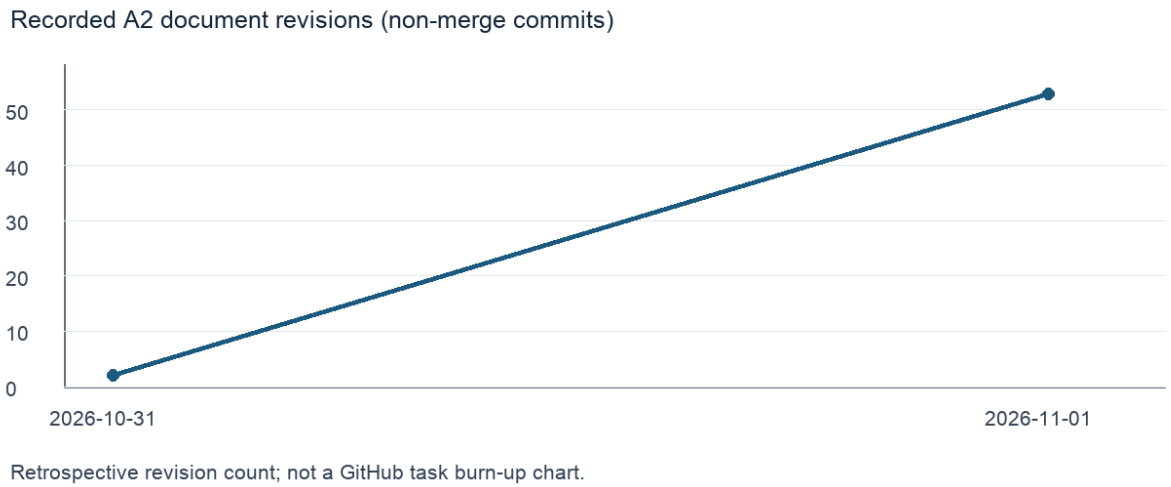


Figure 13.3.1 — Retrospective non-merge commit count for `a2docs`*, from 14 September 2026 to the recorded baseline, using author timestamps. Commits are counted once. This measures document revisions, not original sprint burn-up, task completion or individual effort; author and push times may differ.*

Recorded milestone	Contributor recorded by Git	Evidence
Initial A2 outline added, 31 October	Yousef	f302703
Scope, assumptions, architecture and persistence drafted, 1 November	Yousef	bbff6af; 3d64851; 3b22b3b; dc9cece
Wireframes, third state model and tools discussion integrated, 1 November	Cooper	cbd64c9; 43540f6; c25f4a2
Fault and tracking diagrams/tables integrated, 1 November	Harrison	6893db2; 89d5441; 9bf8a33
Maintenance and references updated, 1 November	Isaac	3de0e75; cde7f15; 9f435fa

Repeated late revisions/merges support earlier integration checkpoints as an improvement. They cannot prove the planned sprint process was followed; original sprint snapshots remain unavailable. Appendix A records the supplied discussion account separately.

14. References and resources

REF-01–REF-05 correspond respectively to SRC-01–SRC-05, whose complete titles, versions and locators are listed in Section 1.3. Additional resources follow.

ID	Resource	Version / location	Use
REF-06	JGraph, diagrams.net / draw.io	draw.io	Architecture and SD-01/SD-02 authoring.
REF-07	Lucid Software, Lucidchart	Lucidchart	Wireframes WF-01–WF-08 and SD-03.
REF-08	Anthropic, Claude	Claude	Assessment-brief clarification, as recorded by the team in its resource list.
REF-09	OpenAI, Codex / ChatGPT	Assistance used during SDD preparation	Drafting and explanation of sections, source comparison, architecture preparation, consistency checks, Git integration support and PDF assembly. This is assistance, not evidence of stakeholder approval, team meetings or executed tests.

ID	Resource	Version / location	Use
REF-10	PostgreSQL Global Development Group	Constraints; transactions; isolation; locking — PostgreSQL 18	Sections 9–10: integrity, transaction boundaries, consistent reads and concurrency.
REF-11	PostgreSQL Global Development Group	Continuous archiving — PostgreSQL 18	Section 9.3 recovery proposal.
REF-12	OWASP Foundation	Authorization Cheat Sheet	Backend permission checks and least privilege.
REF-13	Group 3 Team A	Repository history and individual commit links in Sections 13 and Appendix B	Document contribution and integration evidence.

Appendix A. Team interaction records

A-01

Actual date/time: 31 October 2026, 2:03 am – 1:15 pm
Channel/place: WhatsApp, Group A
Participants/attendance: Yousef, Cooper
Recorder/source: WhatsApp chat export, exported 1 November 2026, 10:18 pm
Agenda/topic: SDD template for Assessment 2

Discussion: Yousef uploaded the SDD template and assigned sections to each member inside it. He said the allocation was not final and anyone could change their part. Cooper acknowledged and said he would review it.

Decisions: Initial section allocation set by Yousef, open to change.

Action	Actual owner	Due date if agreed	Status at the time of the record
Review assigned sections in the SDD template	Cooper	None agreed	Acknowledged

Unresolved points: None **Source link / evidence reference:** EV-01, 31 October section

A-02

Actual date/time: 1 November 2026, 2:41 pm – 2:47 pm
Channel/place: WhatsApp, Group A
Participants/attendance: Cooper, Ayaan, Yousef
Recorder/source: WhatsApp chat export
Agenda/topic: Locating Team B's SRS

Discussion: Cooper could not find Team B's SRS in the forum. Ayaan offered to get it. Yousef said it was in the week 8 Assignment 1 handover forum, and Cooper found it there.

Decisions: None

Action	Actual owner	Due date if agreed	Status at the time of the record
Retrieve Team B SRS	Cooper	None agreed	Done

Unresolved points: None **Source link / evidence reference:** EV-01, 1 November, 2:41–2:47 pm

A-03

Actual date/time: 1 November 2026, 4:29 pm
Channel/place: WhatsApp, Group A
Participants/attendance: Yousef, Harrison
Recorder/source: WhatsApp chat export
Agenda/topic: Progress update

Discussion: Yousef said he had pushed his first section draft. Harrison said he would push his work soon.

Decisions: None

Action	Actual owner	Due date if agreed	Status at the time of the record
Push first section draft	Yousef	None agreed	Done
Push state diagram work	Harrison	None agreed	In progress

Unresolved points: None **Source link / evidence reference:** EV-01, 1 November, 4:29 pm; EV-02; scope draft (document revision; not a push-time record)

A-04

Actual date/time: 1 November 2026, 8:36 pm – 8:56 pm
Channel/place: WhatsApp, Group A
Participants/attendance: Yousef, Harrison, Cooper
Recorder/source: WhatsApp chat export
Agenda/topic: Rebalancing sections and state diagram format

Discussion: Yousef said the sections had been assigned quickly and offered Cooper more work if he had less than the others. Harrison suggested Cooper do the last state diagram (Section 8). Cooper agreed and said he was nearly done with the wireframes. Cooper asked Harrison to push his first two state diagrams so he could follow the same format. Harrison shared the unit's state diagram guide he had used and pushed the first state diagram table.

Decisions: Cooper takes Section 8 (state diagram 3).

Action	Actual owner	Due date if agreed	Status at the time of the record
Complete Section 8 state diagram	Cooper	None agreed	Not started
Push first state diagram table	Harrison	None agreed	Done (8:56 pm)

Unresolved points: None **Source link / evidence reference:** EV-01, 1 November, 8:36–8:56 pm

A-05

Actual date/time: 1 November 2026, 9:03 pm – 9:19 pm
Channel/place: WhatsApp, Group A
Participants/attendance: Harrison, Cooper; Isaac tagged
Recorder/source: WhatsApp chat export
Agenda/topic: Merging work into the main SDD

Discussion: Harrison asked Cooper to move his work into SDD_template. Cooper checked that it was now the main file, and Harrison confirmed that he and Yousef had been working in it. Harrison asked Isaac whether he was doing his parts. Harrison then said he would move his own work over first since he had less.

Decisions: SDD_template is the main working file.

Action	Actual owner	Due date if agreed	Status at the time of the record
Move sections into SDD_template	Cooper, Harrison	None agreed	In progress

Unresolved points: Isaac's progress (no reply until 9:30 pm) **Source link / evidence reference:** EV-01, 1 November, 9:03–9:19 pm

A-06

Actual date/time: 1 November 2026, 9:30 pm
Channel/place: WhatsApp, Group A
Participants/attendance: Harrison, Isaac
Recorder/source: WhatsApp chat export
Agenda/topic: Isaac's sections

Discussion: Harrison asked Cooper whether the work should be split between the other three. Isaac replied that he had been at work and was getting it done.

Decisions: None

Action	Actual owner	Due date if agreed	Status at the time of the record
Complete assigned sections	Isaac	None agreed	In progress

Unresolved points: Whether Isaac's work would be split was not formally decided. **Source link / evidence reference:** EV-01, 1 November, 9:30 pm

A-07

Actual date/time: 1 November 2026, 9:40 pm – 9:48 pm
Channel/place: WhatsApp, Group A
Participants/attendance: Cooper, Harrison
Recorder/source: WhatsApp chat export
Agenda/topic: Basing state diagrams on use cases

Discussion: Cooper said he had pushed the wireframes and was starting the last state diagram. Harrison said the Section 8 header needed to change so the diagram was based on UC-01 View Station Status, because the state diagrams are meant to be based on the use cases. Cooper asked whether the wireframes needed redoing, and Harrison said they did not. Cooper confirmed that SB-01 was already based on UC-01. Harrison said he had changed his last two state diagrams to match the use cases.

Decisions: Section 8 is based on UC-01. The wireframes stay as they are.

Action	Actual owner	Due date if agreed	Status at the time of the record
Push wireframes WF-01 to WF-08	Cooper	None agreed	Done (9:40 pm)
Rewrite Section 8 around UC-01	Cooper	None agreed	In progress

Unresolved points: None **Source link / evidence reference:** EV-01, 1 November, 9:40–9:48 pm

A-08

Actual date/time: 1 November 2026, 9:51 pm – 10:03 pm
Channel/place: WhatsApp, Group A
Participants/attendance: Cooper, Harrison
Recorder/source: WhatsApp chat export
Agenda/topic: Diagram image not showing in the SDD

Discussion: Cooper asked whether Harrison was making a diagram as well as the table. Harrison said he had pushed it, but the image (7chart.png) was in the images folder and not showing in the markdown. Harrison changed the link format, and Cooper then linked the image himself, which fixed it. Harrison said he had nearly finished his second diagram.

Decisions: Cooper fixes the image link.

Action	Actual owner	Due date if agreed	Status at the time of the record
Link 7chart.png in the SDD	Cooper	None agreed	Done (10:02 pm)
Finish second state diagram	Harrison	None agreed	Nearly done

Unresolved points: None **Source link / evidence reference:** EV-01, 1 November, 9:51–10:03 pm

Appendix B. Evidence and contributions

Evidence ID	What it demonstrates	Actual date/period	Source link/file	Related section
EV-01	Team communication, task allocation and decisions	31 Oct – 1 Nov 2026	WhatsApp chat export (Group A)	Appendix A, 13.2
EV-02	Recorded document commits and integration	31 Oct–2 Nov 2026	Repository history	13.2, 13.3, B.2
EV-03	Wireframes WF-01 to WF-08	1 Nov 2026	images/WF-01.png through images/WF-08.png	Section 5
EV-04	State diagrams SD-01 and SD-02	1 Nov 2026	images/7chart.png, images/8chart.png	Sections 6–7
EV-05	State diagram SD-03	1 Nov 2026	images/SD_03.png	Section 8

B.2 Individual contributions supported by evidence

Team member	Work actually contributed	Review/collaboration actually performed	Supporting issue/commit/PR/record
Yousef	Template/allocation; Sections 1, 3, 4, 9.1–9.3, 10, 11 and 13.1; initial 12.1; final integration	Offered to rebalance workload	A-01, A-03, A-04; EV-02
Harrison	Sections 6–7 state models; Section 12 inventory, 26 test cases and test plan; coordinated use-case/state alignment	Shared the state diagram guide; checked Cooper's Section 8 basis	A-03, A-04, A-07, A-08; EV-04
Cooper	Sections 2, 5 and 8; wireframes WF-01–WF-08; final 13.2 tools discussion; Appendices A–B	Fixed the 7chart.png link; helped Isaac with 9.4 and 13.2	A-02, A-04, A-07, A-08; EV-03, EV-05
Isaac	Section 9.4 maintenance; Section 14 reference updates; initial 13.2 contributions	Contributed section updates through GitHub	A-06; EV-02